CASSIA+ Instruction and Plans Booklet

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1 What is the CASSIA model?

1.1 Introduction

The package is available from GitHub: `https://github.com/josimms/CASSIA`

CASSIA is an intra-annual growth model for an individual tree in boreal conditions. The central idea is that each organ has a **phenologically-driven potential growth** rate set by temperature and day length, and an **actual growth** rate determined by how full the tree's carbon stores are:

$$\text{actual growth} = \text{potential growth} \times \text{sugar store fullness} \tag{1}$$

This means carbon limitation is explicit: if photosynthesis is poor, reserves deplete and actual growth falls below its phenological potential even during the growing season. The core mathematical structure is described in Schiestl-Aalto et al. [2015]; this booklet documents the full implementation including extensions added since publication.

CASSIA simulates the following processes:

- **Photosynthesis (GPP):** pre-computed input, PRELES light-use efficiency, or p-hydro hydraulic optimality (Section **??**).
- **Organ growth:** height, diameter, needles, buds, roots, and ectomycorrhiza.
- **Carbon stores:** single aggregate pool (default) or per-organ sugar and starch pools with explicit transfer dynamics (Section **??**).
- **Respiration:** growth and maintenance, temperature-dependent (Section 4).

- **Phenology:** needle cohort ageing and senescence.
- **Xylogenesis:** cell-level early/late-wood formation and ring width.
- **Nitrogen** and **C/N transfer** between tree and ectomycorrhizal fungi *(under development — see below).*

Under Development: Known limitations

- **Nitrogen and C/N transfer:** the co-limitation and mycorrhizal trading sub-models are under active development. Toggles are present but results should be treated as preliminary.
- **p-hydro photosynthesis** (`phydro = TRUE`): the C++ call is not yet wired into the main yearly loop — the model runs but GPP returns zero. Use `preles = TRUE` or the default `photosynthesis_as_input = TRUE` instead.
- **Eco-evolutionary variant** (`ecoevolutionary = TRUE`): the compiled export is currently disabled; passing this toggle will error.
- **Lettosuo and Väriö parameters:** included but not fully calibrated — treat non-Hyde results as indicative only.

Further details for individual functions can be found with `?function_name` in R; a full function list is in Section 7; tutorials are in Section 2.

This booklet also covers newer developments not yet published, including the organ-level sugar model and detailed xylogenesis. Contact the authors with questions about these sections.

1.1.1 Coding plan

The model was originially written in R and then translated to C++. The C++ version of the model changes the running time of the model from 0.397 seconds to 0.006, so is useful for calibrations, but not all subfunctions have been translated. Currently the basic model and the sugar allocation model are the ones that are working. Xylogenesis and water functions will soon be added (email Joanna if needed quicker). In 2024-2025 a soil, mycorrhizal growth and photosynthesis model are being integrated.

All C++ functions should be accessible by an R interface.

1.1.2 Git and GitHub resources

If you are unfamiliar with GitHub then these are some helpful resources. If you would like to develop the model then using GitHub would be ideal, if you just want to use the model then you just need to know how to install the model from GitHub.

- **GitHub:** `https://docs.github.com/en`
- **GitHub:** `https://docs.github.com/en/get-started/quickstart/hello-world`
- **RStudio:** `https://education.rstudio.com/learn/beginner/`

- **R:** `https://www.codecademy.com/learn/learn-r`
- **C++:** `https://www.codecademy.com/learn/learn-c-plus-plus`

2 Tutorials

Interactive tutorials are provided as package vignettes, accessible in R with `browseVignettes("CASSIA")` or by name:

- **Running the Model** (`vignette("Running_The_Model")`) — Install, run a basic simulation with the built-in Hyytiälä weather data, interpret the three output dataframes (`out$Growth`, `out$Sugar`, `out$Preles`), and modify parameters.
- **Visualising Output** (`vignette("Visualisation")`) — Plot weather drivers, compare model output to observations with `plot_comparison`, and inspect organ-level sugar and starch pools.
- **Built-in Datasets** (`vignette("Datasets")`) — Explore and modify the built-in parameter tables (`parameters_p`, `ratios_p`, `sperling_p`, etc.) and reference datasets.
- **Component Models** (`vignette("Component_Models")`) — Use the Repola allometry and PRELES photosynthesis sub-models independently of the main CASSIA loop.
- **Weather Data Preparation** (`vignette("Weather")`) — Three routes to prepare weather input: (1) the built-in `weather_original` dataset, (2) downloading and aggregating SMEAR station data, and (3) processing ERA5 reanalysis data with unit conversions and bias correction.
- **ERA5 Weather** (`vignette("ERA5_Weather")`) — ERA5 reanalysis workflow, the `calculate_VPD` unit-conversion function, and the pre-processed `ERA5_CASSIA_preles` and `ERA5_CASSIA_phydro` package datasets.
- **Ring Width Formation** (`vignette("ring_width_process")`) — Step-by-step walkthrough of how CASSIA computes ring width: temperature response g, cumulative growth state sD, seasonal modulator fD, growth driver GD, early/late-wood cell pools, and potential ring width pot_mm.
- **Sensitivity Analysis** (`vignette("Sensitivity_Analysis")`) — Why and how to run a sensitivity analysis, which parameters matter most for ring-width output, and how to calibrate for a new site.

2.1 Weather Data Format

CASSIA requires a daily weather dataframe. Which columns are needed depends on the photosynthesis sub-model selected; columns not used by the active sub-model may be `NA`.

Always required:

Column	Description	Units	Notes
`dates`	Calendar date	`Date` class	Must be set from year + day-of-year
`X`	Day of year	integer 1–366	
`T`	Air temperature	°C	
`TSA`	Soil temperature, A horizon	°C	≈5 cm depth
`TSB`	Soil temperature, B horizon	°C	≈10 cm depth
`MB`	Volumetric soil moisture, B horizon	$m^3\ m^{-3}$	
`Rain`	Precipitation	mm day^{-1}	

Additional columns by photosynthesis sub-model:

Toggle	Extra columns required	Units
`photosynthesis_as_input = TRUE` (default)	`P` — GPP per tree	g C m^{-2} day^{-1}
`PRELES_GPP = TRUE` (stable) / `preles = TRUE` (dev)	`PAR`, `VPD`, `CO2`, `fAPAR`	mol m^{-2} day^{-1}, kPa, ppm, 0–1
`phydro = TRUE` (dev only)	`PPFD`, `PPFD_max`, `VPD`, `CO2`, `Nitrogen`, `PA`, `SWP`	μmol m^{-2} s^{-1}, μmol m^{-2} s^{-1}, kPa, ppm, g m^{-2}, Pa, MPa

The built-in `weather_original` dataset contains all columns for all sub-models and covers Hyytiälä (SMEAR II) 2015–2017. See `vignette("Weather")` for how to prepare data from a new site.

2.2 Key Functions

The table below lists the functions and datasets most users will interact with directly. A complete reference—including internal helpers, calibration utilities, and all data-processing functions—is given in Appendix 7.

Table 1: Key CASSIA functions and datasets for everyday use.

Function / Dataset	Purpose
	Running the model
`CASSIA_cpp`	Main entry point. Runs the C++ model for one or more years with optional sugar, soil and xylogenesis sub-models. See §8.8 for all toggle arguments.
`initialize_parameters`	Prepare parameter vectors for a given site before calling `CASSIA_cpp`.
`load_data`	Load the Hyytiälä calibration and comparison datasets in one call.
`update_model_settings`	Check that model settings are internally consistent—not all sub-model toggles can be combined.
	Photosynthesis and allometry

Table 1 – continued from previous page

Function / Dataset	Purpose
`PRELES_GPP`	Compute daily GPP from climate data using the PRELES sub-model.
`repola`	Repola (2009) biomass allometry for Scots pine—used internally to initialise organ masses.
	Key datasets
`weather_original`	Daily Hyytiälä SMEAR II weather 2015–2017. See `?weather_original` for column descriptions.
`parameters_p`	Main site-specific parameter matrix (75 rows × 3 sites: Hyde, Lettosuo, Väriö). See Appendix §8.4.
`sperling_p`	Sugar model parameters (55 rows × 3 sites: Hyde, Lettosuo, Flakaliden_c). See Appendix §8.5.
`common_p`	Site-independent parameters (soil physics, gas constants, respiration coefficients). See Appendix §8.2.
`ratios_p`	Allocation ratio parameters (9 rows × 2 sites: Hyde, Lettosuo). See Appendix §8.3.
	Visualisation (representative examples)
`plot_comparison`	Plot model output alongside observational data.
`plot_sugar_starch_comparison`	Compare modelled sugar and starch pools to measurements.
`plot_data`	General-purpose CASSIA output visualisation.
	Weather processing
`calculate_VPD`	Compute VPD from temperature and dew-point temperature (needed for ERA5 inputs).
`downloading_data`	Download SMEAR II weather data for Hyytiälä directly from the SMEAR API.
`whole_weather_process`	End-to-end pipeline: download, aggregate and format raw SMEAR data.

3 CASSIA

3.1 Basic Model

Model Lowdown: CASSIA.

Model type: Carbon based growth model
Inputs: Temperature (air, soil), soil water content, photosynthesis
Scope: Tree level (with categorisation and cell level growth)
Timestep: Daily
Reference: Schiestl-Aalto et al. [2015, 2019]

NOTE: Please check the references [Schiestl-Aalto et al., 2015, 2019] to understand the full structure of the CASSIA model - equations will be added here when water dependencies are added (again email Joanna if these are needed sooner). But the most important aspect is the carbon sink and source effect on seasonal growth:

$$\begin{aligned}\text{Actual Growth} = &\text{Poteital Gorwth Dependant on Environmental Conditions} \cdot \\ &\min\left[\text{Sugar limitation}, \text{Nitrogen Limitation}\right]\end{aligned}$$

Note that the sugar limitation is the original formulation. The nitrogen limitation is currently being added to the system.

3.1.1 Long-Term Height Increment Correction

Over multi-decadal simulations the maximum height and diameter increment rates decline as the increasing structural, hydraulic, and respiratory costs of a larger tree are not fully captured by the short-term growth model. A simple empirical correction reduces the maximum daily increment with tree age A:

$$LH = LH_0 \cdot \exp(-k\,A), \tag{2}$$
$$LD = LD_0 \cdot \exp(-k\,A), \tag{3}$$

where LH_0 and LD_0 are the initial maximum height and diameter increment rates, and k is a calibrated decay constant.

3.2 Sugar Model

CASSIA includes an organ level internal sugar model based on the logic from Sperling et al. [2019], which predicts the bloom dates of almond trees by considering that bloom happens when the sugar level drops beyond a certain threshold. The sugar level is controlled by enzymes that convert sugar to starch. The enzyme activity is affected by the amount of enzymes and temperature. Plant production of enzymes is also generated by the difference between current sugar level and the "expected" sugar level (equilibrium point).

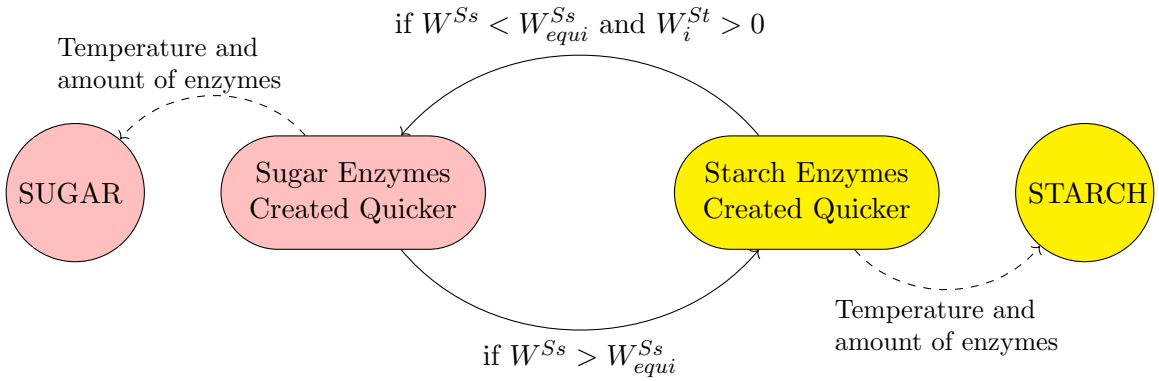


Figure 1: Figure to show the organ-level sugar model process.

This is mathematically represented by;

$$
\begin{aligned}
&W_i^{Ss} > W_{equi,i}^{Ss}: && (4)\\
&\qquad \begin{cases} \frac{dD_i^{Ss}}{dt} = \lambda_i D_i^{Ss} + \delta_i \\ \frac{dD_i^{St}}{dt} = \lambda_i D_i^{St} \end{cases}\\
&W_i^{Ss} < W_{equi,i}^{Ss} \text{ and } W_i^{St} > 0: \\
&\qquad \begin{cases} \frac{dD_i^{Ss}}{dt} = \lambda_i D_i^{Ss} \\ \frac{dD_i^{Sd}}{dt} = \lambda_i D_i^{St} + \delta_i \end{cases}\\
&K_i^{St}(T, D^{St}) = D^{St} e^{F^{St} T} && (5)\\
&K_i^{Ss}(T, D^{Ss}) = D^{Ss} e^{F^{Ss} T} && (6)\\
&Q_i(W^{Ss}) = 0.004211 \rho_i (-K_i^{St}(T, A_i^{St}) + K_i^{Ss}(T, A_i^{Ss})) && (7)
\end{aligned}
$$

Where W is the amount of sugar, Q is the transfer of sugar to starch, Ss is sugar and St is starch, K is the synthesis of of sugar or starch dependent on temperature and amount of relevant enzymes, D is the amount of enzymes, δ, λ, F are all parameters that control the enzymatic behaviour and amount, T is temperature and ρ is the organ density. i determines the organ. Emergency supplies of sugar – when the organ's sugar concentration is less than a threshold L_i – is from the starch storage and is released by $E(W_i^{Ss}, W_i^{St})$ seen in (8).

$$
E\left(W^{Ss}, W^{St}\right) = \begin{cases} \min\left[\max\left[\frac{L_i - W_i^{Ss}}{\tau}, 0\right], W_i^{St}\right] & W_i^{St} \geq 0 \\ 0 & \text{else} \end{cases} \qquad (8)
$$

3.2.1 Winter Sugar Scaling and Physiological Delay

Not all sugar in an organ pool is available for growth and respiration at all times. A minimum sugar concentration must be preserved to support antifreeze protection during winter dormancy [Ögren, 1997, Chang et al., 2021]. The seasonal pattern of this reserve is governed by a smooth

environmental scaling function incorporating a physiological delay that prevents unrealistically abrupt responses to short-term weather changes.

Instantaneous environmental response. The instantaneous winter limitation target $w^*(t)$ is computed from daily air temperature and photosynthetically active radiation:

$$w^*(t) = 1 - f_T(T_{\text{air}}(t)) \cdot f_{\text{PAR}}(\text{PAR}(t)), \tag{9}$$

where $w^* \in [0, 1]$ with values near 1 indicating strong winter limitation. Temperature and light responses are logistic functions:

$$f_T(T_{\text{air}}) = \frac{1}{1 + \exp\left(-\dfrac{T_{\text{air}} - T_{\text{crit}}}{k_T}\right)}, \tag{10}$$

$$f_{\text{PAR}}(\text{PAR}) = \frac{1}{1 + \exp\left(-\dfrac{\text{PAR} - \text{PAR}_{\text{crit}}}{k_{\text{PAR}}}\right)}, \tag{11}$$

with T_{crit}, PAR_{crit} as critical thresholds and k_T, k_{PAR} controlling the sharpness of the transitions. Under warm, bright conditions both functions approach 1 and $w^*(t) \to 0$.

Physiological delay. The realised scaling factor $w(t)$ is updated by a first-order relaxation equation:

$$w(t+1) = w(t) + \frac{1}{\tau}\big(w^*(t) - w(t)\big), \tag{12}$$

where τ (days) is the characteristic acclimation time. This ensures winter limitation responds gradually to sustained changes in temperature and radiation while remaining insensitive to short-lived fluctuations. The system is initialised as $w(1) = w^*(1)$.

3.2.2 Allocation of Sugar Within the Tree

Sugar moves between the organs via a concentration-based model seen in Figure 1. Between each organ and the phloem, there is a diffusion-based relationship [Dietze et al., 2014] seen in Figure 2. The basic relationship is

$$W_{i,j}^{\text{allocation}} = \max\left[(W_i^{Ss} + W_i^{St}) \cdot (f(W_i^{Ss} + W_i^{St}) - f(W_j^{Ss} + W_j^{St})), (W_j^{Ss} + W_j^{St}) - cB_i\right] \tag{13}$$

Where f is a storage term between 0 and 1 which normalizes the concentration gradient. This means that the storage limitation is the driver in the system. This is updated for each iteration, where i is the sugar origin organ, and j, is the destination organ. This is then compared to the storage capacity of each organ cB, where c is the capacity – the percentage of biomass that can be used for storage. These values are fitted with sugar organ level data from Hyytiälä.
As a big debate in the field is the surplus C hypothesis of passive sugar movement compared to the active transfer, both should be considered in terms of their match for the data. Thus, when a carbohydrate conservative strategy is used we use the equation.

$$W_{i,j}^{\text{allocation}} = (W_j^{Ss} + W_j^{St}) - cB_i \tag{14}$$

More allocation strategies could be applied here, however the storage creating a gradient system is quite useful when different allocation strategies are considered in the system (3.2.3).

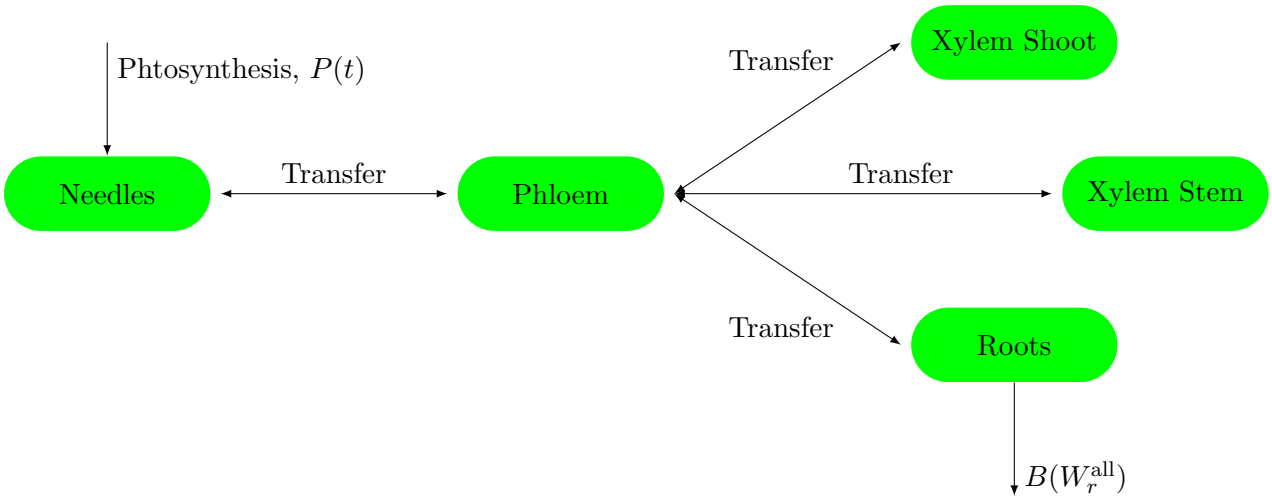


Figure 2: Figure to show the sugar transport of the model

Here $B(W_r^{\text{all}})$ is the allocation to fungal partners or the soil which is determined in the allocation alternatives section.
$G(t)$ is growth associated with that organ and $R(t)$ is both growth and maintenance respiration associated with the organ. Growth and respiration are calculated to be the maximum possible growth [Schiestl-Aalto et al., 2015] and then multiplied by a scaling of the total storage of sugar and starch (f). α, W^{ala} and h are fitted parameters.
Again this f function representing the storage has been formulated in different ways to test the model behaviour. The best ones are the moment are:

$$f(W_i^{all}) = \max\left[0, \min\left[1, h_i\left(1 - \frac{1}{e^{\alpha_i(W_i^{all} - W_i^{ala})}}\right)\right]\right] \tag{15}$$

$$f(W_i^{all}) = \max\left[0, \frac{1}{1 + \exp(-2(\text{Sugar} + \text{Starch} - \text{Storage Capacity}))}\right] \tag{16}$$

Demand-based sugar limitation. An alternative formulation computes the sugar limitation factor $k(t)$ directly from available sugar relative to the organ-specific winter reserve and growth demand:

$$k(t) = \begin{cases} 0 & \text{if } c_k^{Ss} < V_k, \\ \min\left(1, \dfrac{c_k^{Ss} - V_k}{G_k + \varepsilon}\right) & \text{otherwise,} \end{cases} \tag{17}$$

where c_k^{Ss} is the sugar pool (kg C) in organ k, V_k is the organ-specific winter sugar threshold, G_k is the potential growth demand for organ k, and $\varepsilon = 10^{-6}$ prevents division by zero. When available sugar exceeds the growth demand the full potential growth is realised; when sugar falls below the winter threshold no growth occurs.

3.2.3 Allocation to Mycorrhiza / Soil

To understand the observed patterns we should consider a few different strategies. These strategies change the seasonality as well as the magnitude of allocation to the soil.

$$B(W_r^{all}) = \max(0.3 \cdot W_r^{all}, W_r^{all} - cB_r) \tag{18}$$

1. Observed: fitting the data with the sugar available considering a fixed percentage going to the mycorrhiza. This creates the desired effect in the starch. The value of this is approximately 30% of plant sugar being allocated to the soil. No nitrogen limitation.

2. Nitrogen feedback. However, rather than just considering the sugar leaving the system due to it fitting data could it be to harvest nitrogen? In this case consider the difference in growth, which thus effects the carbon fixed in growth and not available to be allocated below ground

3. Nitrogen investment. Again rather than considering this process to be passive, maybe it is active. In this case we consider the plant wanting to minimise the difference between the nitrogen and carbon limitation. Thus investing as much free carbon as it can to minimise this difference (ecoevolutionary inspired)

4. Strategies under development: nitrogen limitation within the model being driven by soil dynamics.

Note that the strategies above are considered due to the data available to us. Eventually the model should consider the ectomycorrizal and plant decisions more explicitly as in 5.2.

3.2.4 Sugar and Starch complete process

Therefore, sugar and starch transfer is calculated via:

$$\frac{dW^{Ss}}{dt} = \begin{cases} P + Q + E - f\,[G + R] - W_{n,p}^{\text{allocation}} & \text{Needles} \\ Q + E - f\,[G + R] - W_{p,s,b,r}^{\text{allocation}} + W_{n,p}^{\text{allocation}} & \text{Phloem} \\ Q + E - f\,[G + R] + W_{p,b}^{\text{allocation}} & \text{Xylem, Shoot} \\ Q + E - f\,[G + R] + W_{p,s}^{\text{allocation}} & \text{Xylem, Stem} \\ Q + E - f\,[G + R] + W_{p,r}^{\text{allocation}} - B(W_r^{all}) & \text{Roots} \end{cases}$$

$$\frac{dW_i^{St}}{dt} = -Q - E$$

The complete organ-level equations in the updated notation are given below. Wood growth (G_w, G_h) is distributed across the phloem, shoot xylem, and stem xylem by mass-fraction weights λ_k (see Section 3.2.6); needle growth G_n and root growth G_r each draw from their own pool. The respiration limitation factor k_k^r caps respiration at zero when sugar falls below 10% of capacity. T_k denotes the transfer flux to organ k from phloem; F_k is the sugar-to-starch net flux (Section 3.2.7); c_s^{Ss} is surplus sugar exported; c_m^{Ss} is sugar allocated to mycorrhiza.

$$\frac{dc_n^{Ss}}{dt} = P(t) - G_n(t)\,k_n(t) - R_n(t)\,k_n^r(t) - T_{\text{phl}} - c_s^{Ss} - F_n \tag{19}$$

$$\frac{dc_{\text{phl}}^{Ss}}{dt} = -\big(G_w(t) + G_h(t)\big)k_{\text{phl}}(t)\,\lambda_{\text{phl}} - \big(R_w(t) + R_h(t)\big)k_{\text{phl}}^r(t)\,\lambda_{\text{phl}} + T_{\text{phl}} - T_{\text{xsh}} - T_{\text{xst}} - T_r - c_m^{Ss} - F_{\text{phl}} \tag{20}$$

$$\frac{dc_{\text{xst}}^{Ss}}{dt} = -\big(G_w(t) + G_h(t)\big)k_{\text{xst}}(t)\,\lambda_{\text{xst}} - \big(R_w(t) + R_h(t)\big)k_{\text{xst}}^r(t)\,\lambda_{\text{xst}} + T_{\text{xst}} - F_{\text{xst}} \tag{21}$$

$$\frac{dc_{\text{xsh}}^{Ss}}{dt} = -\big(G_w(t) + G_h(t)\big)k_{\text{xsh}}(t)\,\lambda_{\text{xsh}} - \big(R_w(t) + R_h(t)\big)k_{\text{xsh}}^r(t)\,\lambda_{\text{xsh}} + T_{\text{xsh}} - F_{\text{xsh}} \tag{22}$$

$$\frac{dc_r^{Ss}}{dt} = -G_r(t)\,k_r(t) - R_r(t)\,k_r^r(t) + T_r - F_r \tag{23}$$

The timing of bud burst is determined by a factor W_{bud}^{Ss} which is close to the minimum possible sugar, when the total storage of sugar goes below this threshold the first day of bud growth begins. The model is run twice for each year, the first run is to find the bud date, and then the model is run again using this value for the bud date to calculate growth. These second run results are the final results of the model. The fact that the equilibrium point and the bud burst have to be defined is an obvious assumption and mean that the system isn't fully dynamic - although the point when the system would reach these hard limits is.
The symbols are defined in Table 2. When the model is calibrared the actual parameter values will be added. Both from literature and from the Bayesian calibration. When the parameters are calibrated, the uniform distribution was chosen due to a lack of prior knowledge.

Table 2: Table to define all of the symbols in the sugar model. TODO: units and values

Symbol	Function	Units (day-1)	Meaning
			Variable
W	Variable	kg C	Amount of of carbon in the organ indicated by the superscript
D	Variable	mg g-1	Amount of enzyme
T	Variable	Degrees C	Temperature, input to the model timeseries
			Process in the model
K	Process	mg g-1	Creation of sugar or starch
Q	Process	kg C	The organ-level sugar model process amalgamated
$E(W^{Ss}, W^{St})$	Process	kg C	Emergency transfer when the sugar concentration is too low if there is enough starch to send an emergency transfer of starch.
$f(W_i^{all})$	Multiplier	0-1	Function between 0 and 1 which scales growth based on the carbohydrate reserves.
$F(W_r^{all})$	Process	kg C	Allocation to mycorrhiza
$R(t)$	Process	kg C	Respiration is worked out as in other CASSIA papers and includes growth and maintenance [Schiestl-Aalto et al., 2019]
$G(t)$	Process	kg C	Growth is worked out as in other CASSIA papers [Schiestl-Aalto et al., 2019]
			Index
Ss	Index		Sugar
St	Index		Starch
all	Index		Both sugar and starch summed
equi	Index		Equilibrium point of relevant variable derived from measurements
n, p, s, b, r	Index		As subscripts these represent all of the organs respectively; needles, phloem, xylem shoot, xylem stem and roots.
			Parameter
λ	Parameter	1e-4	Decay rate of enzymes
δ	Parameter	25e-6	Emzymic parameter
F	Parameter		$= log(Q10)/10$, $Q10$ for sugar synthesis and 1.8 for starch.
0.004211	Parameter		Scale factor from sugar concentration in mg g^{-1} to kg C to match the CASSIA units.
ρ	Parameter		density of organ
L	Parameter		Threshold for appropriate processes
W^{ala}	Parameter		lower bound for storage effects on growth
h	Parameter		Control of the sugar storage effects on growth
α	Parameter		Control of the sugar storage effects on growth
L_r^M	Parameter	kg C	Threshold for allocation to roots
W_{bud}^{Ss}	Parameter	kg C	Threshold of sugar concentration for spring awaken / bud burst
c	Parameter	kg C	The threshold of sugar storage in each organ fitted for each organ with the sugar data collected from Hyytiälä.
			Other
$P(t)$	Timeseries	kg C tree-1	Photosynthesis, timeseries input, calculated by the external model or PRELES depeding on the version of the model. Therefore, there is no dynamic effect on the daily photosynthesis input [Susiluoto et al., 2010].

3.2.5 Code Execution Order

Within each daily time step the model follows an eight-step sequence. Understanding this order is important for interpreting the output and debugging calibrations.

1. **Determine nitrogen capacity.** The nitrogen storage ratio k_k^N is computed for each organ from current and reference nitrogen contents.
2. **Calculate winter sugar limit and ectomycorrhizal strategy.** The organ-specific winter thresholds V_k are updated from the environmental scaling factor $w(t)$, and the active ECM allocation strategy is set.
3. **Apply ectomycorrhizal allocation.** The carbon transferred to ectomycorrhizal fungi c_m^{Ss} is deducted from the phloem pool.
4. **Calculate storage levels.** Current sugar and starch pools are recorded for use in the growth step.
5. **Deduct sugar- and nitrogen-limited growth from pools.** Organ-specific growth demand, downregulated by $k(t)$ and k_k^N, is subtracted from the relevant sugar pools. Growth respiration is included here.
6. **Calculate sink strength to remaining organs.** Demand for roots, xylem (stem and shoot), and phloem is computed from residual storage capacity relative to organ mass bounds.
7. **Needle-to-phloem transfer.** Sugar is transferred from the needle pool to phloem, then distributed from phloem to all other organs and to ectomycorrhiza.
8. **Excess sugar export.** Any needle-pool sugar above the organ storage capacity is exported to the soil as surplus.

3.2.6 Organ Mass Calculation

The sugar storage pools are bounded by the mass of each organ. Needle mass varies within the year following Tian et al. [2021]:

$$B_N = B_N^{\max}\left(\frac{n_{\text{age}}-1}{n_{\text{age}}}\right) + \frac{1}{n_{\text{age}}} B_N^{\max} f_{\text{needle}} f_{\text{sen}}, \tag{24}$$

where $B_N^{\max}$ is the maximum needle biomass (estimated from Repola allometry), n_{age} is the needle cohort lifespan (years), $f_{\text{needle}} \in [0,1]$ reflects the within-season growth increment relative to the maximum needle length from the prior year, and $f_{\text{sen}} \in [0,1]$ represents autumn senescence from the PRELES seasonal modifier.
Sapwood area and mass are derived from needle mass via the pipe model assumption:

$$A_{\text{leaf}} = \frac{B_N^{\text{To Sapwood}}}{\text{LMA}}, \tag{25}$$

$$A_{\text{sapwood}} = \frac{A_{\text{leaf}}}{k_{\text{leaf:sapwood}}}, \tag{26}$$

$$M_{\text{sapwood}} = A_{\text{sapwood}} \cdot H \cdot \rho_{\text{wood}}, \tag{27}$$

where LMA $= 0.086\ \mathrm{kg\,m^{-2}}$ is leaf mass per area [Palmroth and Hari, 2001, Junttila et al., 2022], $k_{\mathrm{leaf:sapwood}} = 1400$ is the leaf-to-sapwood area ratio [Mencuccini and Bonosi, 2001], H is tree height, and $\rho_{\mathrm{wood}} = 200\ \mathrm{kg\,m^{-3}}$ is wood density [Schiestl-Aalto et al., 2019]. Sapwood mass is partitioned into organ pools:

$$M_{\mathrm{xylem,sh}} = \tfrac{1}{9}\, M_{\mathrm{sapwood}}, \tag{28}$$

$$M_{\mathrm{xylem,st}} = \tfrac{8}{9}\, M_{\mathrm{sapwood}}, \tag{29}$$

$$M_{\mathrm{phloem}} = 0.1\, M_{\mathrm{sapwood}}, \tag{30}$$

where the xylem fractions follow Schiestl-Aalto et al. [2019] and the phloem fraction follows Hölttä et al. [2013].

3.2.7 Updated Sugar-Starch Flux Formulation

As an alternative to the enzyme-kinetic Sperling formulation, a smooth, demand-driven flux function computes the net transfer between sugar and starch pools. Positive flux represents sugar $\rightarrow$ starch synthesis; negative flux represents starch $\rightarrow$ sugar remobilisation.
The effective sugar available for starch synthesis, after reserving the winter threshold V, is:

$$s = \max(0,\ c^{Ss} - V), \tag{31}$$

which gates synthesis via a saturating function:

$$\sigma_s = \frac{s}{s + K_s}, \qquad K_s = 0.25\,\Theta, \tag{32}$$

where Θ is the maximum total NSC (non-structural carbohydrate) storage capacity of the organ. The remaining storage capacity limits synthesis:

$$\Theta_{\mathrm{free}} = \max(0,\ \Theta - (c^{Ss} + c^{St})), \qquad \sigma_\Theta = \frac{\Theta_{\mathrm{free}}}{\Theta_{\mathrm{free}} + \varepsilon}. \tag{33}$$

The net flux is then computed as:

$$F_{\mathrm{syn}} = A_0\, \sigma_s\, \sigma_\Theta, \tag{34}$$

$$F_{\mathrm{rem}} = \delta\, c^{St}\, (1 - \sigma_s), \tag{35}$$

$$F = F_{\mathrm{syn}} - F_{\mathrm{rem}}, \tag{36}$$

where A_0 ($\mathrm{day^{-1}}$) is the maximum synthesis rate and δ ($\mathrm{day^{-1}}$) is the remobilisation rate. This formulation ensures synthesis dominates when sugar is abundant, remobilisation dominates when sugar is scarce, and the total NSC pool cannot exceed Θ [Nikinmaa et al., 2013].

3.3 Growth As Investment

Currently, we have considered fluxes and allocation as the key structure. Therefore, the plant investment is in sugars, however we do see a difference in the biomass allocated as well. Currently a framework is being thought of where the objective is to investigate how the cost of different root structures affects an optimisation problem where the cost of photosynthesis is dependent on nitrogen supply.

We seek to:

$$\max\left[\text{Photosynthesis dependent on nitrogen} - \text{Cost of root structures}\right] \tag{37}$$

Which is dependent on number of root tips, length of mycorrhized root tip and allocation of biomass to belowground infrastructure per leaf area. These traits are inspired by studies of the climate and soil fertility gradient in the hemiboreal to arctic soils [Ostonen et al., 2007], the root economic spectrum as explained by Kramer-Walter et al. [2016] and Kong et al. [2019] as well as conversations by boreal root experts.
This formulation has not been published so is not included in this documentation. Contact Joanna for the current state of this work.

3.4 Photosynthesis

In the original CASSIA model used SPP to calculate photosynthesis. The SPP model [TODO: reference] is first run and then the output is added into CASSIA as an input. In terms of integrated photosynthesis models PRELES [Mäkelä et al., 2008, Minunno et al., 2016] is now integrated and phydro [Joshi et al., 2022] with a nitrogen addition is being added.
Preles gives ecosystem level GPP considering environmental conditions and the nitrogen multipler developed by Peltoniemi [TODO: reference] has been added to consider the entire nitrogen porcess in the model. This is explained further below. As PRELES is originally a canopy level model, the output of PRELES has been divided by the amount of trees in Hyyitälä (1010) to give an 'average tree" value. As the amount of trees changes in Hyytiälä, this could also be a variable in the future for Hyyiälä comparisons.
phydro with a nitrogen factor is currently being added to CASSIA to give an eco-evolutionary approach alternative. The nitrogen limitation added will be published in the near future and then added here.

Model Lowdown: PRELES.

Model type: Photosynthesis model with a water balance
Inputs: Climate data, fAPAR
Scope: Stand level
Timestep: Daily
Reference: Mäkelä et al. [2008], Peltoniemi et al. [2015], Minunno et al. [2016]

The current PRELES structure for GPP generation is;

$$\mathrm{P} = \beta \cdot \phi \cdot f_{\mathrm{APAR}} \cdot f_{\mathrm{L}} \cdot f_{\mathrm{S}} \cdot \min\left(f_{\mathrm{D}}, f_{\mathrm{W,P}}\right) \cdot f_{\mathrm{CO_2},P} \tag{38}$$

Where GPP is first defined as the maximum possible under light interception (first two terms) and then down-scaled with daily multipliers for other weather conditions such as light (L), temperature (T), VPD (D), water (W,P) and carbon dioxide (CO2). Therefore to have a nitrogen feedback effect PRELES should be modified to include a f_N multiplier. I suggest this function should be from McMURTRIE [1991]:

$$f_N(N) = \frac{\epsilon}{\epsilon_0} = \frac{\epsilon_0 \left(1 + \frac{N}{N_0}\right) - \sqrt{\left(\epsilon_0 \left(1 + \frac{N}{N_0}\right)\right)^2 - 4\theta\epsilon_0^2 \frac{N}{N_0}}}{2\theta\epsilon_0} \tag{39}$$

Key factors:

- To get the function to be between 0 and 1 divide ϵ with ϵ_0 (the point at the asymptote). Currently N limits at the average needle N concentration (9.1).
- CASSIA used to generate the fAPAR as in Tian et al. [2021]
- When combining CASSIA and PRELES in this way, N is directly limiting photosynthesis, but there are no N controls in CASSIA so growth is only limited indirectly apart from diameter which directly uses GPP.

Calibration data:

- Peltoniemi
- Kainulainen and Holopainen [2002]

Table 3: McMurtrie: table of symbols

Symbol	Function in the system	Units	Meaning
			Variable
N	Variable	Day	Nitrogen
			Parameter
N_0	Parameter	mg g-1	Parameter characterising dependence of ϵ on $[N]$
e_0	Parameter	g MJ-1	Light utilisation coefficient at high $[N]$
θ	Parameter		Curvature parameter of relationship pf ϵ to $[N]$

4 Respiration

Respiration is calculated separately for each organ and has two components: maintenance respiration (proportional to living biomass and temperature-sensitive) and growth respiration (proportional to new growth). Both are subtracted directly from the organ sugar pool each day.

4.1 Maintenance Respiration

Maintenance respiration for organ k follows a Q_{10} formulation:

$$R_m^k(T) = r_k^m \cdot Q_{10,k}^{(T-20)/10} \cdot W_k, \tag{40}$$

where r_k^m is the base respiration rate at 20 °C, $Q_{10,k}$ is the temperature sensitivity, T is air (needles) or soil (roots, wood) temperature, and W_k is the organ biomass. When the sugar pool falls below 10% of storage capacity, maintenance respiration is set to zero to prevent the pool going negative.

4.2 Growth Respiration

Growth respiration is a fixed fraction of the carbon used for new growth:

$$R_g^k = r_k^g \cdot G_k, \tag{41}$$

where r_k^g is the growth-respiration coefficient (see `Rg.N`, `Rg.S`, `Rg.R` in Appendix 10) and G_k is the daily growth of organ k. Growth respiration is included in the sugar deduction at step 5 of the execution order (Section 3.2.5).

Parameter values for all organs are given in Appendix Tables 10 and 12. Below-ground (soil and root) respiration seasonality and its partitioning between autotrophic and heterotrophic components are described empirically in Ryhti et al. [2021, 2022]; a mechanistic coupling of those measurements to the model is in development.

5 Soil Functions (Under Development)

Under Development

The functions in this section are present in the CASSIA package (activated via `soil = TRUE` or `ecoevolutionary = TRUE` in `CASSIA_cpp`) but are under active development. Parameters are not yet fully calibrated for all sites, and equations or function signatures may change between versions. The core CASSIA growth, sugar, and respiration model (Sections 3–5) is the stable, recommended entry point for new users. The Mycofon and Symphony subsections below are the most provisional parts of this section.

The original extended model structure is shown in Figure 3. The equations are then stated afterwords according to the figure's sections.

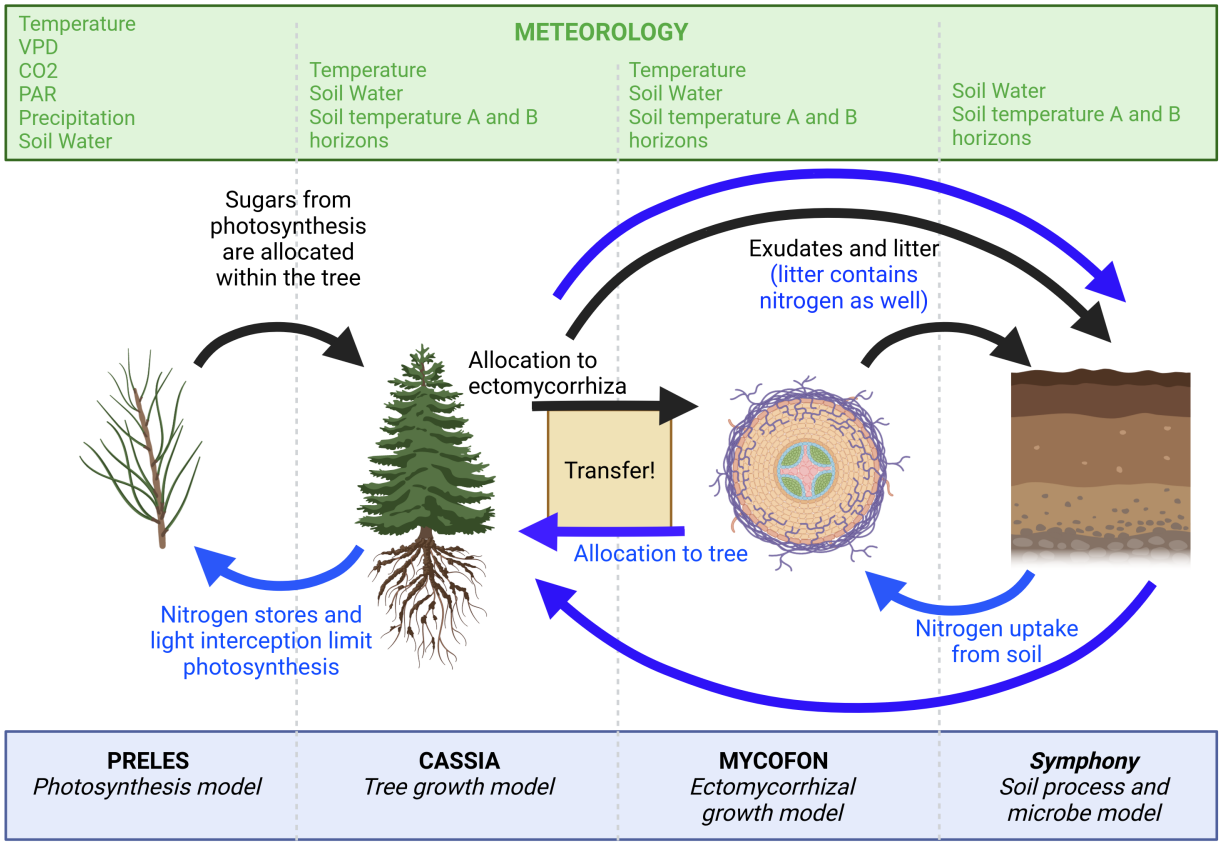


Figure 3: Figure to show the model process. Arrows show the most important inputs to the separate modules in terms of this question. Created with BioRender.com

Assumption:

- No root competition - big problem in a boreal forest! Hyyitälä (the main site in this calibration / model) specifically has been shown to have intense competition [Ryhti et al.,

2022]

- As in Ekblad et al. [2013] assume that the seasonality effect is produced by different parts of the model (N, C inputs and growth)
- The mycorrhiza should be reflective of one tree as there is only C input from one tree - this should naturally limit the growth
- One nitrogen pool for the whole plant - this nitrogen affects the photosynthetic effect, but is not specifically leaf nitrogen nor organ nitrogen for the nitroge limitation on growth
- No spacial considerations in the nitrogen patches. As mycorrhiza and plant roots are observed to create "patches" [Brandes et al., 1998], we can assume that they detect and uptake the nitrogen that is closely available to them and thus they get an average uptake of high and low nitrogen patches. The problem comes when data collecting. The value for the N:C ratio needs to be taken from lots of different locations to make sure that the average is representative of the entire area.
- All nitrogen uptake should be considered - organic, nitrate and ammonium - as they form a significant part of nitrogen uptake in the plant [Wallenda and Read, 1999, Näsholm et al., 2009]

5.1 Uptake Functions

5.1.1 Organ Nitrogen Storage Capacity

Each organ maintains a reference nitrogen concentration derived from empirical measurements. Growth is modulated by the ratio of actual to reference organ nitrogen content:

$$k_k^N = \begin{cases} 1 & \text{if } N_k \geq N_k^{\text{ref}}, \\ \dfrac{N_k}{N_k^{\text{ref}}} & \text{otherwise,} \end{cases} \tag{42}$$

where N_k is the current nitrogen content of organ k and N_k^{ref} is the reference (target) nitrogen content. This capacity term enters the growth equation in parallel with the sugar limitation factor $k(t)$, so that growth is simultaneously constrained by both carbon and nitrogen availability:

$$\text{Actual Growth} = \text{Potential Growth} \cdot \min\big(k(t),\ k_k^N\big)\,. \tag{43}$$

5.1.2 Nitrogen uptake

The organic uptake of N is currently controlled by considering the maximum possible uptake and then this is downscaled with environment dependent functions, taking a value between 0 and 1. These functions take the same logic, but not the same form as the PRELES equations.

$$u(N) = \frac{kN^8}{N_{limit}^8 + N^8} \tag{44}$$

$$f_T(T) = \begin{cases} \frac{T+20}{55} & T > 0 \\ 0 & T \leq 0 \end{cases} \tag{45}$$

$$f_{SWC}(SWC) = \frac{SWC^8}{0.3^8 + SWC^8} \tag{46}$$

$$u_{actual} = f_T(T) f_{SWC}(SWC) u(N) \tag{47}$$

Where fs are the environmental modification functions and $f \in [0, 1]$, $u()$ is the uptake. In equation 44, N is the concentration of a form of nitrogen, N_{limit} and k are parameters to control the uptake. Equation 45 controls the temperature response, where T is the temperature, 20 and 55 are parameters to control the uptake (non-fitted, will change with calibration). Finally the water effect is controlled by Equation 46, where SWC is soil water content and 0.3 is a parameter (will change with calibration). Equation 47 is the final nitrogen uptake effect. The parameters are dependant on the type of nitrogen.

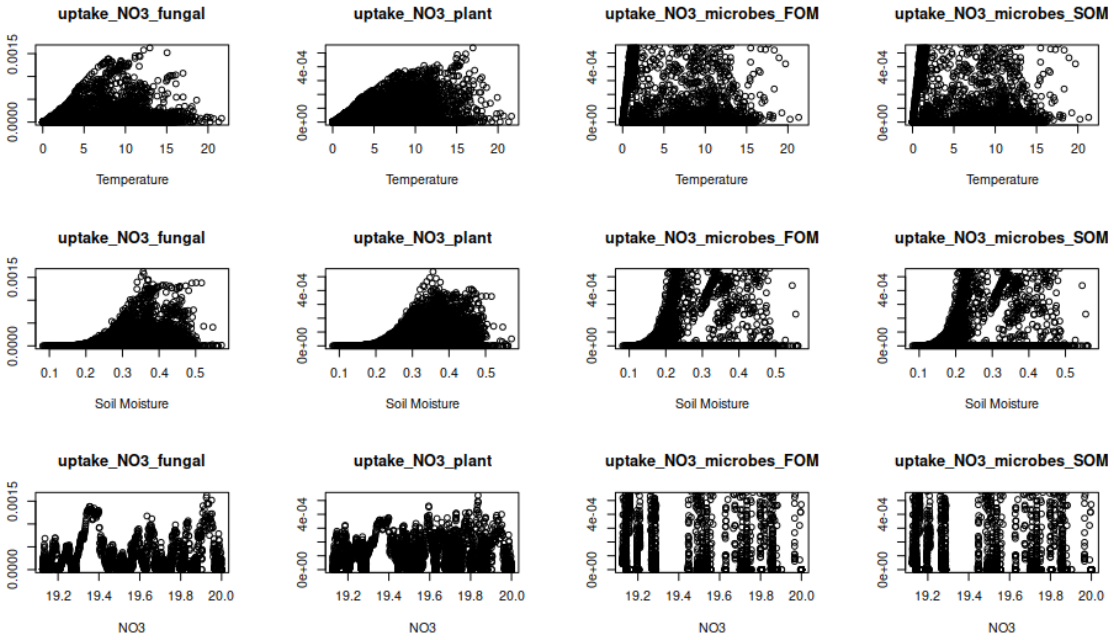


Figure 4: Figure to show the results of the uptake equation with respect to these environmental variables when the simulations are run for a few different years. Note the model is yet to be calibrated so the scales are not sensible.

5.1.3 Microbe Carbon Uptake

This is the carbon uptake of the mycorrhiza and other microbes in the soil. Current version of the model only has microbial uptake. This uptake is formed using the same logic as the nitrogen

uptake. The carbon uptake is in the form of organic compounds, although considered in kg C in the model.

$$u(C) = \frac{kC^8}{C_{\text{limit}}^8 + C^8} \tag{48}$$

$$f_T(T) = \begin{cases} \frac{T+20}{55} & T > 0 \\ 0 & T \leq 0 \end{cases} \tag{49}$$

$$f_{SWC}(SWC) = \frac{SWC^8}{0.3^8 + SWC^8} \tag{50}$$

$$u_{actual} = f_T(T) f_{SWC}(SWC) u(C) \tag{51}$$

5.1.4 Plant N Uptake

NOTE: Uptake of nitrogen can also be considered as proportional to sugar exudated as well as a function of the nitrogen concentration in the soil. The following is considering a case when the soil model or at least nitrogen availability is an input.
This function is for the soil to root uptake only. This function uses the nitrogen uptake functions for each of the individual types of N compiled for the plant. The demand is calculated by the decision function, then used to scale the maximum uptake Section 5.2.1. Although demand is currently 1 for all decision functions included until the code works better and strategies have been chosen that reflect the demand being 1 [Meyer et al., 2010, Franklin et al., 2014]. Eventually, the demand should be inversely proportional to the concentration of the desired nutrient in each organism. The nitrogen transferred to the root is a combination of all of the uptake functions, with the NO3 nitrogen uptake updated with a NH4 modifier. NH4, NO3 and FOM are the amounts in the soil. T is temperature in the soil B horizon and then SWC is the soil water content. For $f(NH4)$, a and b are fitted parameters and the function gives a value between 0 and 1. m is the ratio of mycorrhized roots.
Original formulation:

$$f(NH4) = \frac{aNH4^8}{NH4^8 + b^8} \tag{52}$$

$$\begin{aligned} \text{N to root} = &(1-m) \cdot \text{ demand (plant N amount)} \cdot [u_{organic}(FOM, T, SWC) \\ &+ u_{NH4}(NH4, T, SWC) + f(NH4) u_{NO3}(NO3, T, SWC)] \end{aligned} \tag{53}$$

New formulation: (to be added to code)

$$\text{Allocation factor for roots} = \text{demand (plant N amount)} \tag{54}$$

$$\begin{aligned} \text{N to root} = &(1-m) \cdot [u_{organic}(FOM, T, SWC) \\ &+ u_{NH4}(NH4, T, SWC) + f(NH4) u_{NO3}(NO3, T, SWC)] \end{aligned} \tag{55}$$

More formulations to be added!

Table 4: Uptake functions

Symbol	Function in the system	Units	Meaning
			Variable
N	Variable	kg N	Nitrogen
$NH4$	Variable	kg N	NH4 pool in soil
$NO3$	Variable	kg N	NO3 pool in soil
FOM	Variable	kg N	FOM pool in soil
C	Variable	kg C	Carbon
T	Variable	degree C	Temperature
SWC	Variable	%	Soil water content
m	Variable	0-1	Amount of roots mycorrhized, will eventually by a moving value as in Equation 83
			Parameter
N_{limit}	Parameter	kg N	Parameter that controls when the switch in the uptake is in terms of the N limitations
k	Parameter		Parameter to control the nitrogen
0.3	Parameter	%	Parameter that controls when the switch in the uptake is in terms of SWC
20, 55	Parameter		Parameters controlling the uptake of nitrogen in terms of temperature

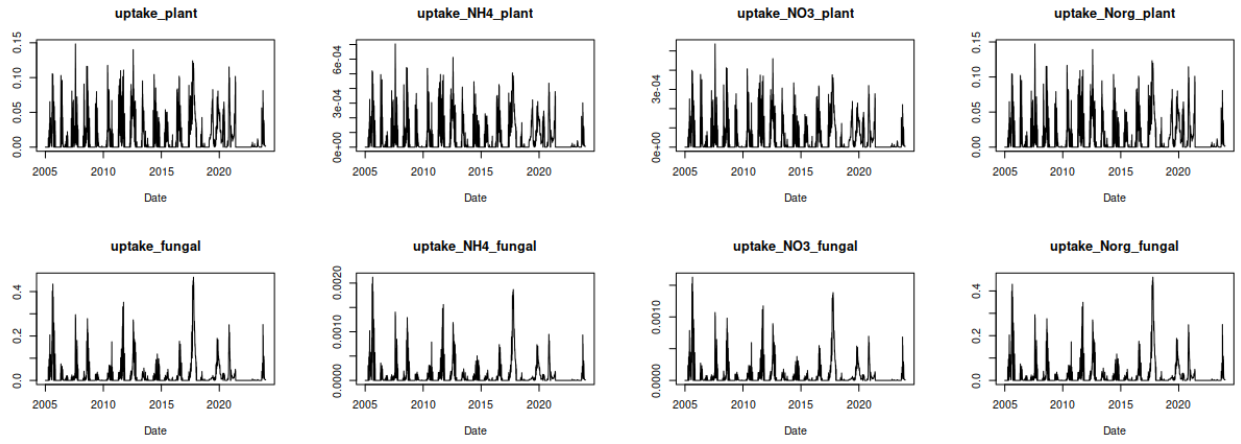


Figure 5: The uptake of nitrogen over the years 2005 to 2023 for the plant and mycorrhiza. Note the model is yet to be calibrated so the scales are not sensible. Behaviour is consistent with literature

Assumptions:

- Almost the same uptake priorities for all of the organisms (trees and mycorrhiza), similar to [Marschner et al., 1991]
- Mineral and organic uptake of N are in parallel due to the fact that when the N forms are found in a mixture the amino acid transfer doesn't decrease in proportion to the inorganic [Näsholm et al., 2009, Wallenda and Read, 1999] and as I could find relationships between NH_4^+ and NO_3^- uptake [Marschner et al., 1991].
- Currently ignoring mass flow: which is bad as Oyewole et al. [2017] notes that "the results suggest that mass flow, induced by transpiration, may be a strong driver for plant nitrogen

acquisition in boreal forests by delivering higher amounts of amino acids and nitrate to plant roots and mycorrhizas." The water effect on growth will be added in the next iteration of the CASSIA model - so could be thought about at this stage. NOTE: Using the uptake as a reference from this paper anyway so the numbers are in the right range, but this could be a mistake if the processes are different to the ones in my model.

5.1.5 Mycorrhizal N Uptake

Like the plant uptake, this function uses the nitrogen uptake functions for each of the individual types of N compiled for the mycorrhiza. The demand is calculated by the decision function, although is currently 1 for all decision functions included [Meyer et al., 2010, Franklin et al., 2014]. Then the output is the N uptake of each of the types of nitrogen calculated in parallel [Wallenda and Read, 1999, Meyer et al., 2010] and then downscaled with the demand function. Original formulation:

$$\begin{aligned}\text{N to mycorrhiza} = \text{Mycorrhizal Biomass} \cdot \text{demand}\,(\text{mycorhiza C amount}) \cdot \\ [u_{organic}(FOM, T, SWC) \\ + u_{NH4}(NH4, T, SWC) + u_{NO3}(NO3, T, SWC)]\end{aligned} \tag{56}$$

New formulation: (to be added to code)

$$\text{Allocation factor for fungal} = \text{demand}\,(\text{mycorhiza C amount}) \tag{57}$$

$$\begin{aligned}\text{N to root} = \text{Mycorrhizal Biomass} \cdot [u_{organic}(FOM, T, SWC) \\ + u_{NH4}(NH4, T, SWC) + f(NH4)u_{NO3}(NO3, T, SWC)]\end{aligned} \tag{58}$$

More formulations to be added!
Where u_{total} is the total mycorrhizal uptake. $u_{organic}$, u_{NO3} and u_{NH4}, are the mycorrhizal uptake of each of the forms of nitrogen. And demand is an input to the function calculated in the ectomycorrhiza decision function 5.2.2.
Assumption:

- Although in the original model there was no mycorrhizal decomposition. Decomposition is currently under development as inspired by [Lindahl et al., 2007, Baskaran et al., 2017]

5.1.6 Sugar Investment and Nitrogen Return

The sugar the tree allocates belowground determines how much nitrogen is returned via two pathways: ectomycorrhizal fungi (mediated by c_m^{Ss}, the sugar sent to mycorrhiza) and free root exudation (mediated by c_s^{Ss}, the surplus sugar exported to soil).
The fraction of total mycorrhizal nitrogen uptake transferred to the tree increases saturatingly with sugar allocation:

$$f_m = \frac{c_m^{Ss}}{c_m^{Ss} + \beta_m}, \tag{59}$$

where β_m is a half-saturation constant encoding the trading behaviour of the fungi.

The fraction of root nitrogen uptake available to the tree decreases as the mycorrhizal allocation increases (because mycorrhizal colonisation inhibits direct root uptake):

$$f_r = \frac{c_s^{Ss}}{c_s^{Ss} + c_m^{Ss} + \varepsilon}. \tag{60}$$

Total daily nitrogen uptake by the tree is then:

$$N_u = f_r \cdot N_r + f_m \cdot N_m, \tag{61}$$

where N_r is root uptake capacity and N_m is mycorrhizal uptake capacity (both from the uptake functions above). The tree's nitrogen balance evolves as:

$$\frac{dN_a}{dt} = N_u - \sum_k G_k \, \gamma_k, \tag{62}$$

where γ_k is the nitrogen-to-carbon ratio of new tissue in organ k.

5.1.7 Microbe Uptake

The microbe uptake holds half of the symphony model dynamics - the microbe side.
Original equations from Symphony paper

$$\text{Carbon Limitation} = u_{\text{Norg}} C_{\text{Microbe}} \tag{63}$$

$$\text{Nitrogen Limitation} = \frac{i(u_{\text{NH4}} + u_{\text{NO3}}) + 0.2 \cdot (\text{N:C})_{\text{Microbes, opt}} C_{\text{Microbe}}}{(\text{N:C})_{\text{Litter}} - (\text{N:C})_{\text{Microbes, opt}}} \tag{64}$$

$$\text{Total N uptake} = \max\{0, \min\{\text{Carbon Limitation}, \text{Nitrogen Limitation}\}\} \tag{65}$$

$$\begin{aligned}\text{Total N uptaken} = &\text{Total N uptake} \cdot \Big(0.2 \cdot (\text{N:C})_{\text{Microbes, opt}} C_{\text{Microbes}} + \\ &(\text{N:C})_{\text{Litter}} - (\text{N:C})_{\text{Microbes, opt}}\Big)\end{aligned} \tag{66}$$

Equations currently implemented

$$\begin{aligned}\text{Total C uptaken} = C_{\text{Microbes}} \, [&u_c(\text{Relevant C compartment}) \\ &+ u_c(\text{Exudes}) - (N:C)_{\text{opt}} + R(T_B)]\end{aligned} \tag{67}$$

$$\text{Total } N_j \text{ Uptaken} = C_{\text{microbe}} \left(u_{N_j} - (\text{N:C})_{\text{Microbes, opt}}\right) \left(\frac{N_j \text{ Pool}}{\text{Total Nitrogen}}\right) \tag{68}$$

$$j \in [NH4, NO3, N_{\text{org}}]$$

Where the uptake functions are the same as earlier, but calibrated for microbe uptake. i is immobilisation, 0.2 is a respiration parameter before the respiration is made dynamic, C, N are carbon and nitrogen respectively and $(N:C)$ are the nitrogen carbon ratios. j represents the NH4, NO3 and organic nitrogen pools. If the SOM decomposes don't get enough nitrogen from their own pools then they can uptake more from the FOM pool, as below.

$$\text{Extra FOM Uptake} = u_{\text{Organic}} \tag{69}$$

Assumptions:

- Carbon and nitrogen have the same units here - need to be careful that they are really made equivalent in the code.
- Exudates are uptaken by the microbes and not by the mycorrhiza. This allows for a limited priming effect.
- In terms of the carbon and nitrogen limitations missing here, they should be present in the soil model – namely immobilisation and growth.

5.2 Decisions: transfer between organisms of N and C

Model Lowdown: Optimisation and Game Theory - currently my own model.

Model type: Decision
Inputs: Max possible transfer of C and N
Scope: Just the transfer amounts of C and N between the organisms
Timestep: Daily
Reference: Meyer et al. [2010], Franklin et al. [2014]

There will be many different strategies here when the model is calibrated and tested as is currently happening in 3.2.3. Plans of these include, but are not limited to:

- Original MYCOFON dynamics [Meyer et al., 2010] *(Coded)*
- Näsholm et al. [2013] strategies *(Coded)*
- The trees / mycorrhiza want to optimise the system, not themselves [Baskaran et al., 2017]
- The trees / mycorrhiza want to optimise their growth (within this C, N and C:N)
- The trees / mycorrhiza want to optimise the a growth form (within this C, N and C:N) [Valverde-Barrantes et al., 2017, Bergmann et al., 2020]
- The trees / mycorrhiza want to optimise their limiting element uptake (within this C, N and C:N)
- The C surplus theory, that C is given proportionally to the overflow from the plant.
- Combinations of these strategies will make the system Game Theoretical

Reference values come from Ingestad et al. [1986], Hobbie [2006], and behaviours from Johnson et al. [2006], Van't Padje et al. [2021], Högberg et al. [2010], Blaško et al. [2015].

5.2.1 Plant Decision

Currently two strategies are coded. These are derived from Näsholm et al. [2013] and the Mycofon model [Meyer et al., 2010]. The strategy of Mycofon gives a demand of 1 and and transfer amount stated in Equation 70.

$$\text{Transfer} = \max\{\text{allo}_{\max} C_r, \text{allo}(C_r - (\text{root} : \text{mycorrhizal})_{\text{opt}}) - C_f\} \tag{70}$$

$$\text{allo}_{\max} = \begin{cases} 1 - (1 - [\exp(-50 * N_r)]^3) & \text{for } N_r < 0.01 \\ 0.2 & \text{else} \end{cases} \tag{71}$$

$$\text{allo} = \begin{cases} \frac{N_{\text{allo}}}{N_r + N_{\text{allo}}} & \text{for } N_{\text{allo}} < 0.5 \\ 1 & \text{else} \end{cases} \tag{72}$$

Where C_r and C_f are the carbon in the roots and mycorrhiza respectively, $(\text{root} : \text{mycorrhizal})_{\text{opt}}$ is the optimal root mycorrhizal biomass ratio, N_r is the nitrogen in the roots and N_{allo} is the nitrogen allocated by the mycorrhiza.
The strategy of Näsholm gives a demand of 1 and and a transfer shown in Equation 73

$$\max\left[C^{\text{CASSIA}}_{\text{Allocated}}, 0.0\right] \tag{73}$$

where CASSIA is the maximum C allocation to the mycorrhiza calculated in CASSIA.

5.2.2 Mycorrhizal Decision

Currently two strategies are used. These are derived from Näsholm et al. [2013] and the Mycofon model [Meyer et al., 2010]. The strategy of Mycofon has a demand of 1 and a transfer function given as below 74.

$$\text{Transfer} = \max\left[N_f \left(1 - \frac{N_r}{C_r \cdot (N : C)_{\text{opt}}}\right), 0.0\right] \tag{74}$$

where N_f, N_r are the N in the mycorrhiza and root respectively, C_r is the C in the root and $(N : C)_{\text{opt}}$ is the optimal N:C ration in the root.
The strategy of Näsholm has a demand of 1 and a transfer of

$$\max\left[N_f - G_f, 0.0\right] \tag{75}$$

Where N and G are the nitrogen and growth of the mycorrhiza.

5.3 Mycorrhizal Code / Mycofon (Under Development)

Under Development: Mycofon

The Mycofon mycorrhizal growth and balance model is implemented in the code but parameters are provisional and the timing of mycorrhizal root growth is currently borrowed from fine-root phenology. The decision and strategy functions (§5.2) are more developed. Treat outputs from this section as qualitative until a dedicated calibration is completed.

5.3.1 Mycorrhizal Growth

The growth of mycorrhiza is controlled currently by the same logic as CASSIA [Schiestl-Aalto et al., 2015], where the mycorrhizal growth is first calculated as possible growth based on temperature (Equation 79). Then the Non-Structural Carbon and Nitrogen form a limitation, as in Meyer et al. [2010] and Franklin et al. [2014]. NOTE: the timing of root growth is used for mycorrhiza temporarily for testing, parameterisation will soon follow.

$$g_F(t) = \begin{cases} 0 & T_a(t) < 0 \\ (1 - \exp(-\lambda M(t))(1 + \exp(-a(T_a(t) - b))^{-1} & T_a(t) \geq 0 \end{cases} \tag{76}$$

Where $g_R(t)$ is the first step in calculating the possible growth. λ is a parameter that decreases fine root growth during water deficiency, $M(t)$ is the soil moisture content, T_a is the temperature at soil depth a, finally a and b are fitted growth parameters.

$$f_F(t) = \begin{cases} 0 & s_i \leq 0 \\ \frac{1}{2}\left(\sin\left(\frac{2\pi}{s_i^c}\left(s_i(t) - \frac{s_i^c}{4}\right)\right) + 1\right) & 0 \leq s_i \leq s_i^c \\ 0 & s_i \geq s_i^c \end{cases} \tag{77}$$

$$G_F = \frac{L0}{\text{Mycelium Lifespan}} \cdot f_F(t) \cdot g_F(t) \tag{78}$$

$$\text{Potential Mycelium Growth} = \begin{cases} G_F & G_F > 0 \\ 0 & G_F \leq 0 \end{cases} \tag{79}$$

More information can be found in Schiestl-Aalto et al. [2015]! NOTE: the rest of the equations for the rest of the organs will be added to their respective sections at a later date. Now the rest of the equations not in CASSIA. The potential ectomycorrhizal growth is then limited by the stores of carbon and nitrogen.

$$f_C \approx \frac{C_{NS}^f}{C^f} \tag{80}$$

$$f_N \approx \frac{N_{NS}^f}{N^f} \tag{81}$$

$$\text{Mycelium Growth} = \text{Potential Mycelium Growth} \cdot f_C \cdot f_N \tag{82}$$

Assumptions:

- Need to assume a upper limit to the storage capacities and the effect that storage would have on growth.

5.3.2 Mycofon Balance

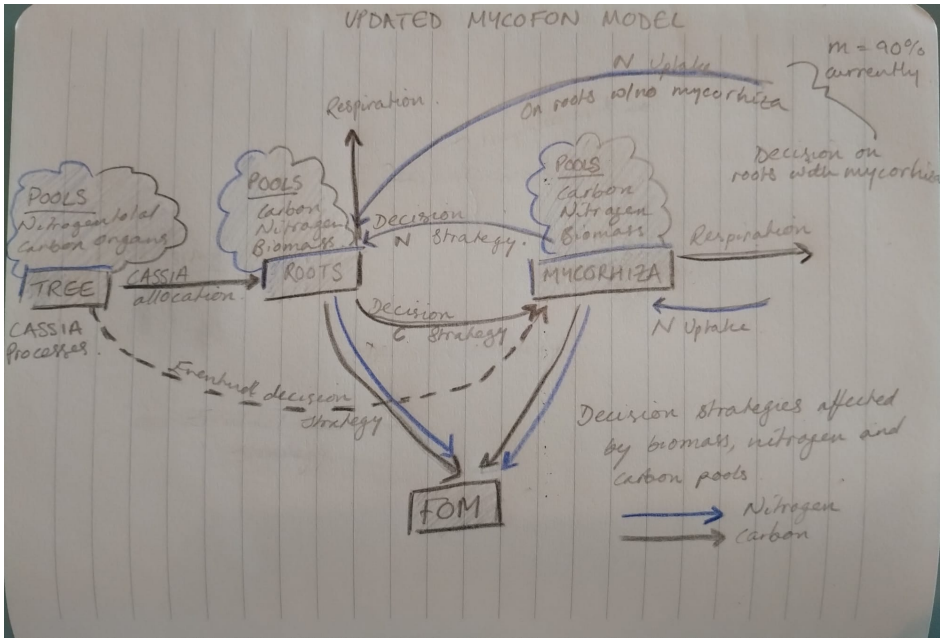


Figure 6: Flow chart for the updated MYOCFON model. Exact processes are represented by an arrow, more detailed processes are represented in the sections / flow diagrams referenced. The uptake rates etc. all have modifications by the environment, which are not written explicitly here.

Model Lowdown: MYCOFON Inspired.

Model type: Daily N and C cycling model, with a little bit of growth
Inputs: Temperature, N and C, biomass
Scope: Tree or stand level, depends on the inputs
Timestep: Daily
Reference: Version inspired by: Meyer et al. [2010]

The official MYCOFON model can be gotten by contacting Hongxing He (hongxing.he@gu.se). Due to the links with the Symphony and CASSIA models I have remade the MYCOFON model in C++ for this project.
The MYCOFON model (Meyer, 2010), with updates to include the uptake properties for three types of nitrogen as well as the decision functionality. This function brings together the C or N balance as in the original MYCOFON model, the decision functions for resource transfer, mycorrhizal growth and multiple N uptake functions. The equations in this function are below.

$$m = \frac{C_f}{B^{rf}_{opt} C^r} \tag{83}$$

$$\frac{dC^r}{dt} = \text{Growth from CASSIA} - (1-m)s^r C^r - m s^m C^r \tag{84}$$

$$\frac{dC^f}{dt} = g^f - s^{\text{mantle}} C^{\text{mantle}} - s^{\text{ERM}} C^{\text{ERM}} - R(T_{\text{B}}) C^f \tag{85}$$

$$\frac{dC^r_{\text{NonStruct}}}{dt} = C^{\text{CASSIA}}_{\text{Allocated}} - d^r(C^{\text{CASSIA}}) \tag{86}$$

$$\frac{dN^r_{\text{NonStruct}}}{dt} = d^f(C^f) + u^f C^f \tag{87}$$

$$\frac{dC^f_{\text{NonStruct}}}{dt} = d^r(C^{\text{CASSIA}}) - g^f \tag{88}$$

$$\frac{dN^f_{\text{NonStruct}}}{dt} = u^f C^f - g^f - d^f(C^f) \tag{89}$$

Table 5: Mycofon: table of symbols

Symbol	Function in the system	Units	Meaning
			Variable
t	Variable	Day	Time
N	Variable	kg N	Amount of nitrogen in specified organs
C	Variable	kg C	Amount of carbon in specified organs
B	Variable	kg C	Biomass
$C^{\text{CASSIA}}_{\text{Allocated}}$	Variable	kg C	The maximum amount of sugar that could be allocated from the CASSIA model
T	Variable / Input time-series	Degrees C	Temperature
m	Variable	0-1	Mycorrhized roots ratio. Although a formula is presented in the document, 0.9 is currently being used for the testing phase.
			Index
r, f	Index		Indexes for the roots and mycorrhiza respectively
opt	Index		Refers to the optimal value of the indicated variable
m, r	Index		Mycorrhized or non-mycorrhized roots
ERM	Index		Index referring to the extra-radical mycelium
Mantle	Index		Index referring to the mantle
B	Index		B soil horizon
			Parameter
s	Parameter	kg C day-1	Turnover
			Function
$d()$	Function	0-1	Decision function, gives the allocated amount of carbon or nitrogen depending on the direction of the function, see Section 5.2
R(Temperature)	Function	kg C	Respiration for indicated compartment, see Section 4
$g()$	Function	kg C	Growth function, for the mycorrhiza

Where N is nitrogen, C is carbon, r roots, f mycorrhiza, B biomass, opt is optimum. The uptake and decision functions are then calculated from the balance of the last iteration. Note that the

plant decision requires the N allocated, so the mycorrhizal decision is calculated first in the code. Where, $C^{\text{CASSIA}}_{\text{Allocated}}$ is the carbon allocated from the CASSIA model, s is turnover, which is different for mycorrhized (m) and non-mycorrhized roots (r), T_B is temperature of the soil, $d()$ is decision (including exudes and transfer to mycorrhiza) and $R()$ is respiration, although for the roots this is calculated in CASSIA not here. Again, the growth process is currently handled in CASSIA so this is just the cumulative sum of the growth with the turnover negated. Growth for mycorrhiza is represented by $g()$.
Note: the m value is currently 0.9, although will be changed to this formulation when the model is fully tested.
Assumptions:

- As the root:mycorrhiza surface area ratio should control the mycorrhizal uptake of C from the tree, and in boreal systems there is a high colonisation rate [Smith and Read, 2010], therefore I assume that there is an optimal relationship one, this assumption could effect the underground dynamics, both in terms of the type of N as well as the colonisation dynamics

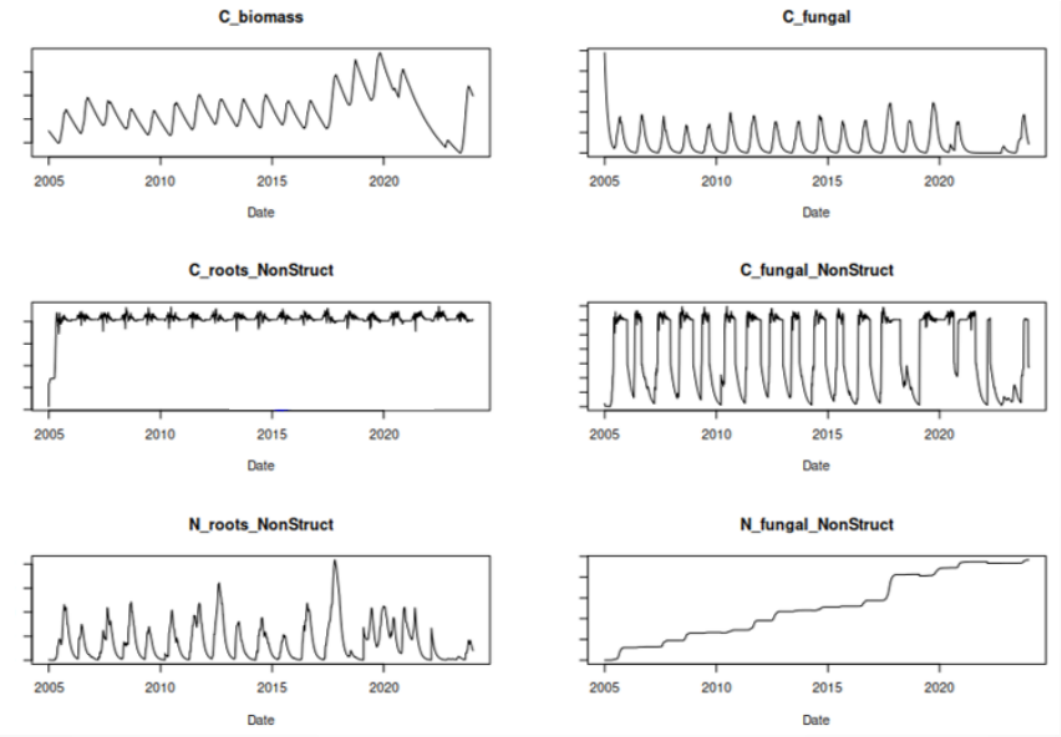


Figure 7: Figure to show the current growth in the model with the Näsholm et al. [2013] decision strategy.

5.4 Soil Processes: Soil Model / Symphony (Under Development)

Under Development: Symphony

The soil C and N cycling model (inspired by Baskaran et al. [2017] and the Symphony model) is functional in the code when `soil = TRUE`, but soil parameters have not been formally calibrated for Hyytiälä or other sites. The equations below reflect the intended model structure; some processes (exudation, ectomycorrhizal decomposition, leaching) are still being added. Do not use quantitative soil outputs from this section in publications without a dedicated calibration.

Whilst the aim of the project is **to account for nitrogen and carbon transfer between trees and ectomycorrhiza in a boreal ecosystem**, to do this a soil model is needed. As the amount of nitrogen in the soil depends on the fungal competition, it doesn't make sense to study the strategic transfer between the tree and their symbionts without the other functional group competition between microbes in the soils. And also the feedback of the competitive advantage this symbiosis gives to the ectomycorrhiza in the soil.

Model Lowdown: Soil Model.

Model type: Soil C and N, Process-Organism Model [Perveen et al., 2014]
Inputs: Litter input, initial soil compartment amounts, temperature of air and soil
Scope: Microbial compartmentalisation of soil, with enzymatic N uptake
Timestep: Daily
Reference: Perveen et al. [2014] Baskaran et al. [2017]

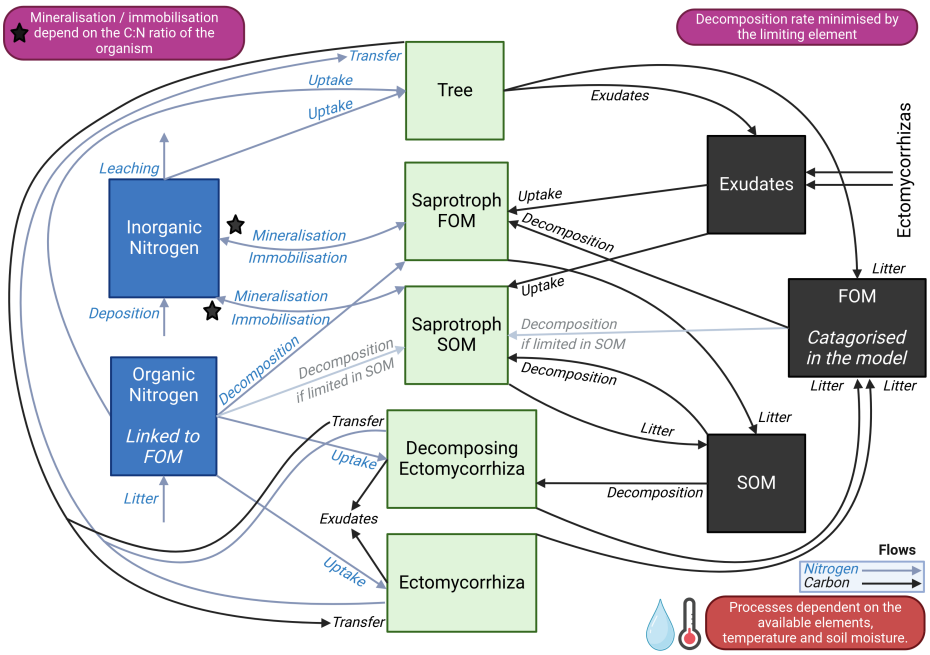


Figure 8: Temporary figure for the soil model. Figure has all of the processes that we want to be represented in the equations.

As the model is seasonal and conceptual so the model should be looked at at two levels: Short term results:

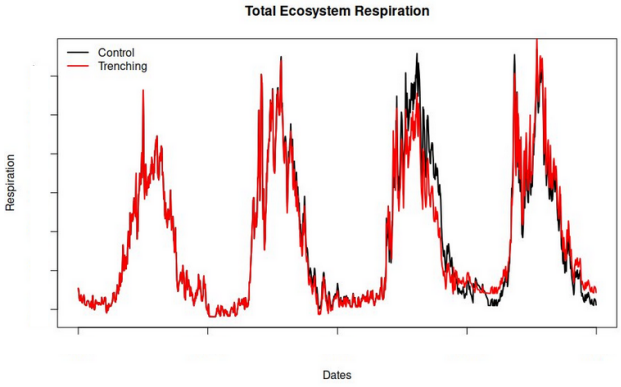
- Priming effect from trenching experiments *(priming effect can be seen)* [Ryhti et al., 2021, 2022]
- Nitrogen and Carbon transfer tracked at Hyytiälä (data being gathered)
- Growth patterns within the seasonal pattern [Ekblad et al., 2013] (roots come from the tree model [Ding et al., 2020] and nitrogen availability based root strategy being added)

The long term results should respond to disturbance:

- Change in fungal composition / dominance of fungal strategy Lindahl et al. [2021]
- Changes in carbon stock after disturbances reflecting the different recycling strategies of the different microbes Clemmensen et al. [2015], Sterkenburg et al. [2015]
- Nitrogen immobilisation in old growth forests Sterkenburg et al. [2015]
- Inorganic vs organic nitrogen stock changes after disturbances Lindahl et al. [2010]

This soil model was originally inspired by Perveen et al. [2014], but in order to make the model fit for boreal forests the model has ended up looking a lot like Baskaran et al. [2017] and to a lesser extent [Sulman et al., 2014]. Only with a seasonal timestep and some small differences in the equations. This is encouraging in terms of the understanding of the system, however should be

carefully considered in terms of the model use. Exploring using another soil model is also in progress.
The behaviour so far:



$$\frac{dC_{\text{SOM}}}{dt} = -u_{\text{C}}(C^{SOM})C^{\text{microbe, SOM}} + s\left(C^{\text{microbe, SOM}} + C^{\text{microbe, FOM}}\right) \tag{90}$$

$$C_{FOM} = C_{FOM}^{needles} + C_{FOM}^{woody} + C_{FOM}^{roots} + C_{FOM}^{mantle} + C_{FOM}^{ERM} \tag{91}$$

$$\begin{aligned} N_{FOM} =& N_{FOM}^{needles}(N:C)^{needles} + N_{FOM}^{woody}(N:C)^{woody} \\ &+ N_{FOM}^{roots}(N:C)^{roots} + N_{FOM}^{mantle}(N:C)^{mantle} + N_{FOM}^{ERM}(N:C)^{ERM} \end{aligned} \tag{92}$$

$$d_{total} = u_C(C^{FOM})C^{microbes,SOM} + u_C(C^{FOM})C^{microbes,FOM} \tag{93}$$

$$\frac{dC_{FOM}^{needles}}{dt} = L^{needles} - d_{total}\frac{C_{FOM}^{needles}}{C_{FOM}} \tag{94}$$

$$\frac{dC_{FOM}^{woody}}{dt} = L^{woody} - d_{total}\frac{C_{FOM}^{woody}}{C_{FOM}} \tag{95}$$

$$\frac{dC_{FOM}^{roots}}{dt} = L^{roots} - d_{total}\frac{C_{FOM}^{roots}}{C_{FOM}} \tag{96}$$

$$\frac{dC_{FOM}^{mantle}}{dt} = L^{mantle} - d_{total}\frac{C_{FOM}^{mantle}}{C_{FOM}} \tag{97}$$

$$\frac{dC_{FOM}^{ERM}}{dt} = L^{ERM} - d_{total}\frac{C_{FOM}^{ERM}}{C_{FOM}} \tag{98}$$

Where C and N are carbon and nitrogen respectively, $(N:C)$ is the carbon nitrogen ratio of the indicated compartment, L is letter and the compartments are refereed to explicitly. FOM is fresh organic matter and SOM is soil organic matter. Where d is the decomposition, which is the uptake of both microbe pools, which is seen in more detail in Section 5.1.3. NOTE: the uptake is only from the aggregated FOM pool rather than each of the separate FOM pools and the decomposition from each pool is currently based on the size of the pool rather than the type of material. This will be changed to reflect the type of material.

$$\frac{dNH4}{dt} = (\psi_{\text{ims}} + \psi_{\text{imf}}) \frac{NH4}{NH4 + NO3} - NH4^{p}_{used} - NH4^{f}_{used} - u_{NH4} C^{microbes,FOM} - u_{NH4} C^{microbes,SOM} \tag{99}$$

$$\frac{dNO3}{dt} = (\psi_{\text{ims}} + \psi_{\text{imf}}) \frac{NH4}{NH4 + NO3} - NO3^{p}_{used} - NO3^{f}_{used} - u_{NO3} C^{microbes,FOM} - u_{NO3} C^{microbes,SOM} \tag{100}$$

$$\frac{dN^{FOM}}{dt} = \text{Litter} - N^{FOM,p}_{used} - N^{FOM,f}_{used} - u_{Norg} C^{microbes,FOM} - u_{Norg} C^{microbes,SOM} \tag{101}$$

$$\frac{dN^{SOM}}{dt} = \text{Microbe Litter} - u_{Norg} C^{microbes,SOM} \tag{102}$$

Where $NH4^{p}_{used}$ is the NH4 used for the plant, f represents the mycorrhizal uptake. This is an input to the function from the mycofon model. Next the microbe balance is considered.

$$\frac{dC^{\text{microbe, FOM}}}{dt} = u_{FOM} C^{\text{microbe, FOM}} - r(T_B) - s \tag{103}$$

$$\frac{dC^{\text{microbe, SOM}}}{dt} = u_{SOM} C^{\text{microbe, SOM}} - r(T_B) - s \tag{104}$$

Assumptions and behaviours:

- Microbes assumed to be organisms that are not in symbiosis with the tree and that form competition for the nitrogen resources from the same pools as trees or ectomycorrhiza.
- Therefore the priming effect is still included as if there is more C in the soil then the decomposition will increase to a N storage bound value. The next day the N uptake can compensate for the N that was used the previous day with the C increase. This means that over time there would be a priming effect - but it is not instantaneous.
- In the model there are no chemical processes - as the idea is to look at symbionts and microbes this make sense, but could be a bad assumption.
- The respiration is not variable yet. Ryhti et al. [2022] data and models planning to be used for this.

Table 6: Soil Model: table of symbols

Symbol	Function in the system	Units	Meaning
			Variable
t	Variable	Day	Time
N	Variable	kg N	Amount of nitrogen in specified organs
C	Variable	kg C	Amount of carbon in specified organs
T	Variable	Degrees C	Temperature
NH4	Variable	kg N	The amount of NH4 in the soil
NO3	Variable	kg N	The amount of NO3 in the soil
			Index
needles	Index		Subscript indicating the needles
woody	Index		Subscript indicating the woody biomass
roots	Index		Subscript indicating the roots
ERM	Index		Subscript indicating the extra-radical mycelium
mantle	Index		Subscript indicating the mantle
FOM	Index		Subscript indicating the fresh organic matter
SOM	Index		Subscript indicating the soil organic matter
p, f	Index		Subscripts indicating the plant and fungi respectively
B	Index		Soil B horison
NH4	Index		For processes related to NH4 pools
NO3	Index		For processes related to NO3 pools
Norg	Index		For processes related to Norg pools
microbe	Index		For processes related to microbes
			Parameter
s	Parameter	kg C day-1	Turnover
			Function
u	Function	kg N	Uptake for indicated compartment, see section 5.1.3

6 Complete table of variables

Table 7: All Sections: Table of Symbols

Symbol	Function in the System	Units	Meaning
			Variable
W	Variable		Amount of carbon in the indicated organ
A	Variable		Amount of enzyme
T	Variable	°C	Temperature (input to the model timeseries)
t	Variable	Day	Time
N	Variable		Amount of nitrogen in specified organs
C	Variable		Amount of carbon in specified organs
B	Variable	kg C	Biomass
$C^{\text{CASSIA}}_{\text{Allocated}}$	Variable		Maximum sugar allocatable from the CASSIA model
m	Variable	0–1	Mycorrhized roots ratio (0.9 used in testing phase)
NH_4	Variable		Amount of NH_4 in soil

NO_3	Variable		Amount of NO_3 in soil
			Index
r, f	Index		Roots and mycorrhiza indices
opt	Index		Refers to the optimal value of the variable
m, r	Index		Mycorrhized or non-mycorrhized roots
ERM	Index		Extra-radical mycelium
Mantle	Index		Mantle tissue
$needles$	Index		Needles
$woody$	Index		Woody biomass
$roots$	Index		Roots
ERM	Index		Extra-radical mycelium
$mantle$	Index		Mantle
FOM	Index		Fresh organic matter
SOM	Index		Soil organic matter
p, f	Index		Plant and fungi
B	Index		Soil B horizon
NH_4	Index		Related to NH_4 pools
NO_3	Index		Related to NO_3 pools
Norg	Index		Related to organic nitrogen pools
Microbe	Index		Related to microbial processes
			Parameter
λ	Parameter		Decay rate of enzymes
δ	Parameter		Enzymatic parameter
A	Parameter		Rate of carbohydrate production
B	Parameter		$= \log(Q_{10})/10$, with Q_{10} for sugar synthesis and 1.8 for starch
0.004211	Parameter		Scale factor from mg g^{-1} sugar concentration to kg C
ρ	Parameter		Density of organ tissue
L	Parameter		Threshold for certain processes
W_{ala}	Parameter		Lower bound for storage effects on growth
h	Parameter		Control parameter for storage effects on growth
α	Parameter		Another control parameter for storage effects on growth
L_r^M	Parameter		Threshold for allocation to roots
W_b^{Ss}	Parameter		Threshold of sugar concentration for spring awakening / bud burst
s	Parameter	kg C day^{-1}	Turnover rate
			Function
$d()$	Function	0–1	Decision function allocating C or N depending on context
$u()$	Function		Uptake for specified compartment (see Section 5.1.3)
R(Temperature)	Function		Respiration function (see Section 4)
$g()$	Function	kg C	Growth function for mycorrhiza
			Process

K	Process	Creation of sugar or starch
Q	Process	Organ-level sugar model amalgamated process
$E(W^{Ss}, W^{St})$	Process	Emergency transfer of starch to sugar under low sugar conditions
$f(W_i^{all})$	Multiplier	Scaling function (0–1) for growth based on carbohydrate reserves
$B(W_r^{all})$	Process	Allocation to mycorrhiza
		Other
$P(t)$	Timeseries	Photosynthesis (input timeseries; no feedback)
$G(t)$	Input, state variable	Growth based on CASSIA model [Schiestl-Aalto et al., 2019]
$R(t)$	State variable	Respiration including growth and maintenance [Schiestl-Aalto et al., 2019]
D	Data	Observational data from Hyytiälä
CASSIA(θ)	Model	CASSIA model with Hyytiälä-specific inputs and parameters

7 Appendix: Complete Function and Dataset List

This table lists every function, dataset and utility in the CASSIA package. For the key user-facing subset, see §2.2.

Table 8: All CASSIA package functions and datasets.

Function / Dataset	Purpose	Doc'd
	Tests and validation	
all_tests	Test the C++ CASSIA function against Pauliina's original R version.	
bound_checks	Check parameter bounds.	
preles_test	Test the C++ PRELES implementation.	
respiration_test_cpp	Test the C++ respiration formulation.	
setup_export_data	Create dataframes for CASSIA model outputs (not yet used).	N
tests	General test runner.	
update_model_settings	Check that model settings are internally consistent—not all toggles can be combined.	N
validate_site	Verify that parameters correspond to a site defined in the model.	N
validate_weather_data	Check that weather data are within allowable bounds.	N
	Plots	
plot_comparison	Plot model output alongside observational data.	
plot_data	General-purpose CASSIA output visualisation.	
plot_mycofon_data	Visualise Mycofon sub-model outputs.	
plot_nitrogen_uptake	Plot nitrogen uptake model output.	
plot_rad_comparison	Compare modelled and observed radiation.	
plot_respiration_data	Visualise respiration components.	
plot_soil_data	Plot soil sub-model outputs.	
plot_sugar_starch_comparison	Compare modelled sugar and starch pools to measurements.	
plot_total_ecosystem_respiration	Plot total ecosystem respiration.	
plot_weather_variables	Visualise weather driver time series.	
respiration_plot	Diagnostic plot for the C++ respiration formulation.	N
xylem_plot	Diagnostic plot for xylogenesis_wrapper.	N
	C++ model functions	
CASSIA_cpp	Main entry point. Runs the C++ model for one or more years with optional sub-models.	Y
CASSIA_calibration	Calibration wrapper around CASSIA_cpp.	

Continued on next page

Table 8 – continued from previous page

Function / Dataset	Purpose	Doc'd
CASSIA_soil	Low-level C++ runner for the soil-coupled variant. Called by `CASSIA_cpp` when `soil = TRUE`.	
CASSIA_yearly	Low-level C++ runner for the standard annual model. Called internally by `CASSIA_cpp`.	Y
Fungal_N_Uptake	C++ fungal nitrogen uptake function.	
Microbe_Uptake	C++ microbial carbon uptake function.	
myco_decision	Mycorrhizal allocation decision function.	
myco_growth	Mycorrhizal growth sub-model.	
mycofon_balence	Mycofon carbon/nitrogen balance calculator.	
plant_decision	Plant-side allocation decision function.	
Plant_N_Uptake	C++ plant nitrogen uptake function.	
uptake_C	Carbon uptake helper (C++).	
uptake_N	Nitrogen uptake helper (C++).	
symphony	Symphony soil model (C++).	
symphony_plus	Extended Symphony soil model (C++).	
symphony_plus_daily	Daily-stepping Symphony variant (C++).	
xylogenesis_wrapper	C++ wrapper for the xylogenesis sub-model.	
	R model functions	
PRELES_GPP	PRELES photosynthesis model in R. Required for the pure-R CASSIA variant.	Y
repola	Repola (2009) biomass allometry for Scots pine. Required for organ mass initialisation.	Y
	Weather processing	
calculate_VPD	Compute VPD from temperature and dew-point temperature. Needed for ERA5 inputs.	Y
downloading_data	Download SMEAR II weather data for Hyytiälä from the SMEAR API.	Y
downloading_rad_data	Download radiation data from the SMEAR API.	
ERAS_reading_nc	Read ERA5 NetCDF files for use in CASSIA.	
fill_missing_values	Interpolate or fill gaps in weather time series.	
generating_max_values_monthly	Compute monthly maxima from sub-daily data.	N
generating_mean_values_daily	Compute daily means from sub-daily data.	N
generating_mean_values_monthly	Compute monthly means from sub-daily data.	N
generating_sum_values_monthly	Compute monthly sums from sub-daily data.	N
Hyytiala_Data_Creation	Assemble the Hyytiälä weather dataset from raw files.	
inporting_hyytiala_raw_data_into_list	Read raw Hyytiälä observation files into a list.	
process_files	Process downloaded SMEAR weather files.	
process_files_co2	Process downloaded SMEAR CO_2 files.	
process_weather_data	General weather data processing pipeline.	

Continued on next page

Table 8 – continued from previous page

Function / Dataset	Purpose	Doc'd
raw_to_daily_monthly_hyytiala	Aggregate raw Hyytiälä data to daily and monthly resolution.	
rbinddatatable	Row-bind a list of data.tables efficiently.	
read_and_combine_weather_data	Read and combine multiple weather files into a single data frame.	
replace_value_DataFrame	Replace a value in a data frame by column and row name.	
simplifying_names	Standardise column names across weather data sources.	
sort_CMIP	Sort pre-downloaded CMIP6 data for Hyytiälä.	
soil_against_yearly	Compare soil model outputs to annual observations.	
whole_weather_process	End-to-end pipeline: download, aggregate and format raw SMEAR data.	
	Datasets	
common_p	Site-independent parameters (soil physics, gas constants, respiration coefficients). See §8.2.	Y
data_format	Hyytiälä data in CASSIA input format (internal use).	
GPP_2009	Internal reference GPP data for 2009.	
GPP_previous_sum	Internal lagged GPP accumulator.	
GPP_ref	Reference daily GPP vector (365 values) for Hyytiälä. Used when `LN_estim = TRUE`.	Y
Hyde_daily_original	Daily Hyytiälä observations for model comparison.	
Hyde_weather	Internal Hyytiälä weather object.	
Hyde_weather_CASSIA	Hyytiälä weather pre-formatted for CASSIA.	
Hyde_yearly_original	Annual Hyytiälä observations for model comparison.	
karike_df_all	Internal calibration dataset.	
leap_years	Logical vector of which years in the weather data are leap years.	
nitrogen	Internal nitrogen input dataset.	
oyewole_2015_calibration_data	Calibration data from Oyewole et al. [2017].	
parameters_p	Main site-specific parameter matrix (75 rows × 3 sites: Hyde, Lettosuo, Väriö). See §8.4.	Y
parameters_R	Soil and nitrogen sub-model parameters (numeric vector, length 58). See Appendix.	Y
ratios_p	Allocation ratio parameters (9 rows × 2 sites: Hyde, Lettosuo). See §8.3.	Y
repo_p	Repola allometry parameters. See Appendix.	Y
sperling_p	Sugar model parameters (55 rows × 3 sites: Hyde, Lettosuo, Flakaliden_c). See §8.5.	Y
weather_original	Daily Hyytiälä SMEAR II weather 2015–2017. See `?weather_original`.	Y

Continued on next page

Table 8 – continued from previous page

Function / Dataset	Purpose	Doc'd
weather_original_2015	Weather data for Hyytiälä 2015.	
weather_original_2016	Weather data for Hyytiälä 2016.	
weather_original_2017	Weather data for Hyytiälä 2017.	
weather_original_2018	Weather data for Hyytiälä 2018.	
	Calibration	
dhtn	Alternative distribution function for boreal calibration.	N
likelyhood_nitrogen_uptake	Likelihood function for the nitrogen uptake sub-model.	N
likelyhood_soil_models	Likelihood function for the soil sub-models.	N
likelyhood_sugar_model	Likelihood function for the sugar sub-model.	N
nitrogen_uptake_calibration	Nitrogen uptake calibration wrapper.	
soil_model_calibration	Soil model calibration wrapper.	
	Internal helpers	
daily_list	Internal Hyytiälä data processing helper.	
daily_to_average	CMIP6 data daily-to-average helper.	
growth_wrapper	Internal growth calculation wrapper.	
initialize_parameters	Prepare parameter vectors for a site.	
printColumnNames	C++ debugging utility—print column names of an Rcpp DataFrame.	
transform_variable	CMIP6 variable transformation helper.	

8 Appendix: Parameter Values in Code

8.1 repo_p Explanation

These parameters are from the paper Repola [2009] and is a multi-variable model fitted for Scots Pine and Norway Spruce in Finland.

Parameter Name	Parameter Value	Equations
b0.repo	-6.303	This model is from Repola [2009] and not explained further here. As the model is a mixed effect model, their parameters here do not have clear biological meanings, but are fixed effect parameters.
b1.repo	14.472	
b2.repo	-3.976	
uk.repo	0.109	
eki.repo	0.118	

Table 9: Explanation for the repo_p parameters built into the package.

8.2 common_p Explanation

Parameter Name (Name in Paper	Parameter value	Units	Comments and additional references
a (a)	0.185	Degree C^{-1}	Parameter of function g
b (b)	18.4	Degree C	Parameter of function g
TR0	0		Root temperature factor
abs_zero	273.15	K	Absolute zero temperature
b.s	4.14		Soil water potential and conductance
theetta.FC	0.62		Hölttä et al. 2009 Table 2
phi.e	-6.8×10^{-7}	MPa	Hölttä et al. 2009 Table 2, note psi.s should be psi.e
K.sat	24.5	mol m^{-1} s^{-1} MPa^{-1}	Hölttä et al 2009, Table 2
R.length	5300	m root m^{-2}	Index, Hölttä et al 2009
M.H2O	0.018	kg mol^{-1}	Hölttä et al 2009
r.cyl	4.25×10^{-3}	m	Hölttä et al 2009
r.root	3.0×10^{-3}	m	Hölttä et al 2009
ypsilon	1×10^{-14}		To prevent dividing by zero
Rg.N (r_R^g)	0.35	kg C(kg C)$^{-1}$	Growth respiration, share of growth (Needles)
Rg.S (r_{wood}^g)	0.3	kg C(kg C)$^{-1}$	Growth respiration, share of growth (Wood)
Rg.R (r_N^g)	0.35	kg C(kg C)$^{-1}$	Growth respiration, share of growth (Fine roots)
gas.const (R)	8.314	J mol^{-1} K^{-1}	Gas constant
M.C (M_C)	12.01	g mol^{-1}	Molar mass of Carbon
M.H	1.008	g mol^{-1}	Molar mass of Hydrogen
M.O	16	g mol^{-1}	Molar mass of Oxygen
osmotic.sugar.conc	2000000	Pa	Osmotic sugar concentration 2 MPa Hölttä et al. 2000
m_n	0.02174605		
Uggla	1.95	–	Division to early/late wood

Table 10: Explanation for the common_p parameters built into the package

8.3 ratio_p Explanation

Parameter Name (Name in Paper)	Parameter value (in package: Hyde, Lettosuo only)			Units	Comments and additional references
	Hyde	Lettosuo	Väriö (ref)	Chinese Site (ref)	
form_factor (φ)	0.6	0.6	0.55		Ratio Lettosuo: The multiplier between a cylinder (with diam=D0, height=h0) and the total biomass of stem, coarse roots and branches
needle_fineroot_ratio	NA	1/2.9	NA		Ratio Depends on tree size, species and site and *
sapwood.share	0.8	0.8	0.8		Ratio
height_growth_coefficient (α_S)	4.3	1	2.6		– repeated value, Variö: leader shoot length / average measurement shoot length (average

8.4 parameters_p Explanation

Table 12: Explanation for the parameters_p vector built into the package.

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
		Respiration			
Q10.N (q_{N}^{10})	1.898	1.898	1.898	–	Needles Q10
Rm.NR (r_N^m)	0.00267	0.0020	0.004005	kg C(kg C)$^{-1}$	Needles R0
Q10.S (q_{Wood}^{10})	1.74788	1.74788	1.74788	–	Wood Q10
Rm.S (r_S^m)	5.5576e-5	5.5576e-5	8.3364e-5	kg C(kg C)$^{-1}$	Wood R0
Q10.R (q_{R}^{10})	2.5575	2.9662	2.0244	–	Fine roots Q10
Rm.R (r_R^m)	0.00958	0.0059	0.00945	kg C(kg C)$^{-1}$	Fine roots R0
		Growth			
sRc (S_R^c)	30	30	25	–	Root growth cessation
sR0 (S_R^0)	-2.5	-2.5	-3	–	Root growth start offset
		Mycorrhiza			
growth.myco	0.1	0.1	0.1		Mycorrhiza growth
root.lifetime	1.7	2.01	2	years	Root lifetime
		Shoots			
HH0	10	10	10	mm	Initial shoot length
sH0	-1.359200388	-4.12008	-1.44	–	Shoot growth beginning
LH	8.226401284	8.226401284	9.0		Shoot growth rate

Continued on next page

Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
LH0	8.226401284	8.226401284	9.0		Initial shoot growth rate
sHc	14.59636279	12.8328	10.0	–	Shoot growth cessation
		Needles			
sN0 (S_N^0)	-8.37584	-3.56589	-5.6	–	Needle growth beginning
TODO: the first LN parameter is a typo, it should be LN0, but will solve this issue later!					
LN	1.849493	1.849493	3.5	mm d^{-1}	Needle growth rate
LN0 (L_{N0})	1.849493	1.849493	3.5	mm d^{-1}	Initial needle growth rate
sNc (S_N^c)	5.263883	7.60671	4.327	–	Needle growth cessation
HN0	1	1	1	mm	Initial needle length
		Diameter			
sD0.Trad	-3.724083738	-3.5	-2.434161213	–	Diameter growth beginning
TODO: the first LD parameter is a typo, it should be LD0, but will solve this issue later!					
LD	1.293443902	1.9	2.9	cells d^{-1}	Diameter growth rate
LD0 (L_{D0})	1.293443902	1.9	2.9	cells d^{-1}	Initial diameter growth rate
sDc (S_D^c)	5.077004992	5.2	4.093829285	–	Diameter growth cessation

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
sDc.T.count	NA	8.8	NA	–	Diameter growth cessation count
Duration parameters					
tau.Ee (τ_e^{early})	10.68685877	5.5	10.25174759	day	Early wood cell enlargement duration
tau.El (τ_e^{late})	8.789131263	4.8	4.510400352	day	Late wood cell enlargement duration
tau.We (τ_{wa}^{early})	25.29448857	17.8	51.60724145	day	Early wood cell wall formation duration
tau.Wl (τ_{wa}^{late})	35.12148687	19.2	17.76015932	day	Late wood cell wall formation duration
GPP					
tau.GPP (τ_{GPP})	5	5	5	–	GPP effect on daily LD
Uggla	1.95	1.8	1.95	–	Division to early/late wood
Buds					
sB0	171	171	181	day	Bud growth beginning
sBc	85	85	60	day	Bud growth cessation
LB	0.005	0.005	0.003		Bud growth rate
Xylogenesis					
cell.d.ew (d_{cell}^{early})	35.7e-6	32.1e-6	30e-6	m	Early wood cell diameter

Continued on next page

Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
cell.d.lw ($d_{\text{cell}}^{\text{late}}$)	24.2e-6	27.5e-6	20e-6	m	Late wood cell diameter
cell.l.ew ($l_{\text{cell}}^{\text{early}}$)	2.59e-3	2.89e-3	2.59e-3	m	Early wood cell length
cell.l.lw ($l_{\text{cell}}^{\text{late}}$)	2.73e-3	2.97e-3	2.73e-3	m	Late wood cell length
cell.wall.density.ew	57	570	557	kg C m^{-3}	Early wood cell wall density
cell.wall.density.lw	57	680	557	kg C m^{-3}	Late wood cell wall density
wall.thickness.ew	2.61e-6	3.1e-6	2.61e-6	m	Early wood wall thickness
wall.thickness.lw	5.23e-6	3.88e-6	5.23e-6	m	Late wood wall thickness
cell.volume.growth.per.day.ew	NA	5.49e-13	NA	m^3 day^{-1}	Early wood cell volume growth rate
cell.volume.growth.per.day.lw	NA	4.62e-13	NA	m^3 day^{-1}	Late wood cell volume growth rate
density_tree (ρ)	400	400	400	kg m^{-3}	Tree density
carbon_share	0.5	0.5	0.5	kg kg^{-1}	Carbon share
D0 (d_0)	0.175	0.175	0.154	m	Initial diameter
h0	17.9	17.9	9.5	m	Initial height
n_age	3	5	5	years	Needle lifespan
n_lenght	34.241	13	39.2	mm	Average needle length

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
h_increment	309.0938	309.0938	120.00	mm	Mean height increment
SLA	13	5.5	13	m^2 kg^{-1}	Specific leaf area
LR0 (L_R^0)	0.07446064	NA	0.02	kg C d^{-1}	LR0 parameter
Repola parameters					
b0_repo	-6.303	NA	NA	–	Repola [2009] Parameter
b1_repo	14.472	NA	NA	–	Repola [2009] Parameter
b2_repo	-3.976	NA	NA	–	Repola [2009] Parameter
Sperling parameters					
lower_bound_needles	0.02	NA	NA	kg C	Lower bound for needles
lower_bound_phloem	0.03	NA	NA	kg C	Lower bound for phloem
lower_bound_roots	0.05	NA	NA	kg C	Lower bound for roots
lower_bound_xylem_sh	0.03	NA	NA	kg C	Lower bound for shoot xylem
lower_bound_xylem_st	0.1	NA	NA	kg C	Lower bound for stem xylem
tau_emergancy_needles	3	NA	NA	–	Emergency time constant for needles

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
tau_emergancy_phloem	3	NA	NA	–	Emergency time constant for phloem
tau_emergancy_roots	3	NA	NA	–	Emergency time constant for roots
tau_emergancy_xylem_sh	3	NA	NA	–	Emergency time constant for shoot xylem
tau_emergancy_xylem_st	3	NA	NA	–	Emergency time constant for stem xylem
lower_bound_W	0.01	NA	NA	kg C	Lower bound for W
tau_emergancy	3	NA	NA	–	General emergency time constant
uk_repo	0.109	NA	0.109	–	Repola [2009] Parameter
eki_repo	0.118	NA	0.118	–	Repola [2009] Parameter
stem_no	3	NA	NA	–	Repola [2009] Parameter
diameter_start_day	79	NA	NA	day	Day of year when diameter growth starts

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
xylem_start_estimate	4.5	NA	NA	mm	Initial estimate for xylem ring width
GPP_mean	463.88	NA	NA	$g\,C\,m^{-2}\,yr^{-1}$	Mean annual GPP used to normalise needle and xylem growth
GPP_initial	481.3	NA	NA	$g\,C\,m^{-2}\,yr^{-1}$	Initial annual GPP estimate for first simulation year
Pipe-Model Corrections (Structural Balance)					
tau_H (τ_H)	5	NA	NA	–	Mäkelä [1999] Eqn. 28 damping exponent; `height_diameter_correcti` RSS-fit against real SMEAR height/diameter (1998–2022 Hyde run)

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
tau_R (τ_R)	5	5	5	–	Mäkelä [1999] Eqn. 28 damping exponent; `root_leaf_correction`, also `root_turnover`'s litter boost. Not independently calibrated, defaults to tau_H
tau_N (τ_N)	5	5	5	–	Mäkelä [1999] Eqn. 28 damping exponent; `leaf_mass_correction`. Not independently calibrated, defaults to tau_H/tau_R
sLc_litter	2.163312	NA	NA	–	Width of the f_S^2-based needle litter bell curve; `leaf_mass_correction`. Fitted against 2006–2016 Hyytiälä litter-trap data, not independently calibrated

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
Litter Nitrogen Content & Retranslocation					
theta_N_needles ($\theta_{needles}$)	4.9	NA	NA	mg N g^{-1}	Korhonen et al. [2013] Table 2: needle litter N concentration, Hyytiälä SMEAR II, 2006–2008
theta_N_roots (θ_{roots})	7.7	NA	NA	mg N g^{-1}	Helmisaari et al. [2007] Fine root N concentration, Scots pine stand near SMEAR II, quoted in Korhonen et al. [2013] §4.2
theta_N_myco (θ_{myco})	3.0	3.0	3.0	mg N g^{-1}	Placeholder (no literature source found); reuses theta_N_roots
retrans_needles ($r_{needles}$)	0.73	NA	NA	–	Korhonen et al. [2013] §3.6.2–3.6.3: needle+leaf retranslocation, 73% of initial foliage N, Hyytiälä

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Table 12 continued from previous page

Parameter Name (Name in Paper)	Parameter value			Units	Comments
	Hyde	Lettosuo	Väriö		
retrans_roots (r_{roots})	0.2	0.2	0.2	–	Wojciechowska et al. [2020] Fine-root N concentration decline, control to fully senescent, *Populus trichocarpa* (not this site/species)
retrans_myco (r_{myco})	0.0	0.0	0.0	–	No resorption assumed; no clean biological analogue for a fungal symbiont's own shed tissue

Note on these parameters. The pipe-model correction exponents and litter-nitrogen rows above were added 2026-11-09, moved into `parameters_p` from being separate `CASSIA_cpp()` arguments so that function's only non-toggle arguments are the big parameter/common/etc. tables. None are independently calibrated for Lettosuo/Väriö; see `data-raw/building_data.R` for the full sourcing notes behind each value.

8.5 sperling_p Explanation

Table 13: Explanation for the sperling_p vector built into the package

Parameter Name	Parameter value			Units	Comments
	Hyde	Lettosuo	Flakaliden_c		
starch0	0.3246781	0.40	0.40		Initial starch concentration
sugar0	0.4184208	0.35	0.35		Initial sugar concentration
starch.needles0	0.03	NA	NA		Initial needle starch concentration
starch.phloem0	0.037	NA	NA		Initial phloem starch concentration
starch.xylem.sh0	0.034	NA	NA		Initial shoot xylem starch concentration
starch.xylem.st0	0.166	NA	NA		Initial stem xylem starch concentration
starch.roots0	0.057	NA	NA		Initial root starch concentration
sugar.needles0	0.087	NA	NA		Initial needle sugar concentration
sugar.phloem0	0.27	NA	NA		Initial phloem sugar concentration
sugar.roots0	0.014	NA	NA		Initial root sugar concentration
sugar.xylem.sh0	0.0249	NA	NA		Initial shoot xylem sugar concentration
sugar.xylem.st0	0.021	NA	NA		Initial stem xylem sugar concentration
Wala	0.0	0.0	0.0		General Wala parameter
Wala.needles	0.0	0.0	0.0		Needle-specific Wala parameter
Wala.phloem	0.0	0.0	0.0		Phloem-specific Wala parameter
Wala.xylem.sh	0.0	0.0	0.0		Shoot xylem Wala parameter
Wala.xylem.st	0.0	0.0	0.0		Stem xylem Wala parameter
Wala.roots	0.0	0.0	0.0		Root-specific Wala parameter

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Table 13 continued from previous page

Parameter Name	Parameter value			Units	Comments
	Hyde	Lettosuo	Flakaliden_c		
carbon.sugar	0.4211	0.4211	NA		Carbon content in sugar
carbon.starch	0.4444	0.4444	NA		Carbon content in starch
alfa	3	3	3		General alfa parameter
alfa.needles	3	3	3		Needle-specific alfa parameter
alfa.phloem	3	3	3		Phloem-specific alfa parameter
alfa.xylem.sh	3	3	3		Shoot xylem alfa parameter
alfa.xylem.st	3	3	3		Stem xylem alfa parameter
alfa.roots	3	3	3		Root-specific alfa parameter
tau.s	2	2	2		Tau s parameter
tau.t	2	2	2		Tau t parameter
starch00	0.3246781	0.40	0.40		Secondary initial starch concentration
sugar00	0.4184208	0.35	0.35		Secondary initial sugar concentration
Q10s	3	NA	NA		Q10 for synthesis
Q10d	1.8	NA	NA		Q10 for decomposition
SCb	0.23	NA	NA	kg C	Storage carbon baseline
sugar.level	0.41	NA	0.35	kg C	Sugar level at senescence
Ad0.needles	0.017	NA	NA		Initial needle Ad0
Ad0.phloem	0.008	NA	NA		Initial phloem Ad0
Ad0.roots	2e-04	NA	NA		Initial root Ad0
Ad0.xylem.sh	2e-04	NA	NA		Initial shoot xylem Ad0
Ad0.xylem.st	0.047	NA	NA		Initial stem xylem Ad0
lamda.needles	0.197	NA	NA		Needle lamda parameter
lamda.phloem	0.05301	NA	NA		Phloem lamda parameter
lamda.roots	0.211	NA	NA		Root lamda parameter
lamda.xylem.sh	0.00401	NA	NA		Shoot xylem lamda parameter
lamda.xylem.st	0.00401	NA	NA		Stem xylem lamda parameter

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Table 13 continued from previous page

Parameter Name	Parameter value			Units	Comments
	Hyde	Lettosuo	Flakaliden_c		
delta.needles	0.729	NA	NA		Needle delta parameter
delta.phloem	0.832	NA	NA		Phloem delta parameter
delta.roots	0.853	NA	NA		Root delta parameter
delta.xylem.sh	0.762	NA	NA		Shoot xylem delta parameter
delta.xylem.st	0.294	NA	NA		Stem xylem delta parameter
percentage_needle_storage	0.11	NA	NA	–	Fraction of needle carbon available as accessible storage
percentage_phloem_storage	0.11	NA	NA	–	Fraction of phloem carbon available as accessible storage
percentage_xylem_sh_storage	0.11	NA	NA	–	Fraction of shoot xylem carbon available as accessible storage
percentage_xylem_st_storage	0.11	NA	NA	–	Fraction of stem xylem carbon available as accessible storage
percentage_roots_storage	0.11	NA	NA	–	Fraction of root carbon available as accessible storage
myco.thresh	0.025	NA	NA		Mycorrhiza threshold

8.6 parameters_R Explanation

`parameters_R` is a numeric vector of length 58 containing parameters for the soil and mycorrhizal sub-models. It is passed internally when `CASSIA_cpp` is called with `soil = TRUE` or `ecoevolutionary = TRUE`. Parameters are indexed positionally (not by name). The table below lists the currently implemented positions; remaining entries are reserved for future soil model development.

Index	Value (Hyde)	Description
1	0.0169	Nitrogen uptake k parameter for organic N
2	200	Nitrogen uptake N_{limit} for organic N
3	120	Nitrogen uptake N_{limit} for NH4
4	1	Nitrogen uptake k for NH4
5	0.5	SWC limit for nitrogen uptake
...	(remaining 53 entries: soil C/N pool parameters, fungal uptake kinetics, microbe parameters)	
58	—	(reserved)

Table 14: Selected entries from the `parameters_R` vector. Full parameter descriptions are maintained in `data-raw/Mycorrhizal_Model_Parameters.R`.

8.7 GPP_ref Explanation

`GPP_ref` is a named numeric vector of length 365 containing the mean daily gross primary production (g C m^{-2} d^{-1}) for each day of year at Hyytiälä, derived from multi-year SMEAR II eddy-covariance measurements. It is used internally by `CASSIA_cpp` when `LN_estim = TRUE` to scale needle elongation by the ratio of the current year's July–August GPP to the long-term reference. The values are built by `data-raw/Hyytiala.R`.

8.8 `CASSIA_cpp` Toggle Reference

The `CASSIA_cpp` function controls which sub-models are active via logical toggle arguments. The table below maps each toggle to the relevant section of this booklet and notes the associated parameter inputs.

Table 15: `CASSIA_cpp` toggle arguments and the model components they activate.

Toggle argument	Model section	Description
	Sugar / Carbon Allocation	
`organ_level_sugar`	Sugar Model	Enables organ-level sugar/starch allocation with per-organ concentration pools (Equations 4–7). Associated parameters: `sperling_p`, `ratio_sugar`, `tau.*` time constants.
`surplus_c`	Sugar Model, §3.2.3	Switches sugar transport from concentration-gradient diffusion to the passive carbon-surplus strategy.
	Soil / Mycorrhiza	
`soil`	Soil Functions	Activates MYCOFON-inspired soil C/N cycling: microbial decomposition, nitrogen uptake functions, and ectomycorrhizal C/N balance. Calls `CASSIA_soil` internally. Parameters: `parameters_R_in` (see Appendix).
`ecoevolutionary`	Decisions (§5.2)	Enables the eco-evolutionary plant–mycorrhiza decision strategy (nitrogen-investment, minimising N/C limitation difference).
	Photosynthesis	
`preles`	Photosynthesis	Uses PRELES to compute daily GPP internally rather than accepting GPP as input. Parameters: `pPREL` (length-32 vector).
`phydro`	Photosynthesis	Replaces PRELES with the p-hydro eco-evolutionary photosynthesis model. Parameters: `phydro_param`.
`LN_estim`	Appendix (`GPP_ref`)	Scales needle elongation by the ratio of the current year's peak GPP to the long-term Hyytiälä reference `GPP_ref`.
`fAPAR_Tian`	Photosynthesis	Derives fAPAR from CASSIA LAI output using the Tian et al. (2021) disaggregation method, feeding the value into PRELES.
	Nitrogen	

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Toggle argument	Model section	Description
`nitrogen_balance`	Photosynthesis	Applies the McMurtrie f_N multiplier to PRELES GPP, so nitrogen concentration directly limits photosynthesis.
`nitrogen_change`	Photosynthesis	Activates a nitrogen deposition change scenario. The nitrogen input is shifted by `nitrogen_contrast` (kg N ha^{-1} yr^{-1}) after the year specified in `driver_N`.
	Environment & Diagnostics	
`environment_effect_xylogenesis`	Basic Model	Enables soil temperature and water content modulation of xylogenesis timing.
`etmodel`	Photosynthesis	Enables the Penman-Monteith evapotranspiration model instead of the default ET formulation.
`LOGFLAG`	–	Writes verbose C++ diagnostic output to the console. For development and debugging only.

Sugar transport time constants. When `organ_level_sugar = TRUE`, sugar movement between organs (Section 3.2.3) is governed by organ-pair diffusion time constants (in days), passed via `sperling_p`:

- `tau.t.needles` — needles ↔ phloem
- `tau.t.phloem` — phloem (hub, outward)
- `tau.t.roots` — phloem ↔ roots
- `tau.t.xylem.sh` — phloem ↔ xylem shoot
- `tau.t.xylem.st` — phloem ↔ xylem stem
- `tau.myco` — roots → mycorrhiza / soil (see Equation 1)

The `ratio_sugar` argument (length-4 numeric vector) sets the initial concentration ratios for sugar transport between the four non-phloem organs.

References

Preetisri Baskaran, Riitta Hyvönen, S Linnea Berglund, Karina E Clemmensen, Göran I Ågren, Björn D Lindahl, and Stefano Manzoni. Modelling the influence of ectomycorrhizal decomposition on plant nutrition and soil carbon sequestration in boreal forest ecosystems. *New Phytologist*, 213(3):1452–1465, 2017.

Joana Bergmann, Alexandra Weigelt, Fons van Der Plas, Daniel C Laughlin, Thom W Kuyper, Nathaly Guerrero-Ramirez, Oscar J Valverde-Barrantes, Helge Bruelheide, Grégoire T Freschet, Colleen M Iversen, et al. The fungal collaboration gradient dominates the root economics space in plants. *Science Advances*, 6(27):eaba3756, 2020.

Róbert Blaško, Lisbet Holm Bach, Stephanie A Yarwood, Susan E Trumbore, Peter Högberg, and Mona N Högberg. Shifts in soil microbial community structure, nitrogen cycling and the concomitant declining n availability in ageing primary boreal forest ecosystems. *Soil Biology and Biochemistry*, 91:200–211, 2015.

Bettina Brandes, Douglas L Godbold, Arnd J Kuhn, and Georg Jentschke. Nitrogen and phosphorus acquisition by the mycelium of the ectomycorrhizal fungus paxillus involutus and its effect on host nutrition. *The New Phytologist*, 140(4):735–743, 1998.

Christine Yao-Yun Chang, Katharina Bräutigam, Norman PA Hüner, and Ingo Ensminger. Champions of winter survival: cold acclimation and molecular regulation of cold hardiness in evergreen conifers. *New Phytologist*, 229(2):675–691, 2021.

Karina E Clemmensen, Roger D Finlay, Anders Dahlberg, Jan Stenlid, David A Wardle, and Björn D Lindahl. Carbon sequestration is related to mycorrhizal fungal community shifts during long-term succession in boreal forests. *New Phytologist*, 205(4):1525–1536, 2015.

Michael C Dietze, Anna Sala, Mariah S Carbone, Claudia I Czimczik, Joshua A Mantooth, Andrew D Richardson, and Rodrigo Vargas. Nonstructural carbon in woody plants. *Annual review of plant biology*, 65(1):667–687, 2014.

Yiyang Ding, Pauliina Schiestl-Aalto, Heljä-Sisko Helmisaari, Naoki Makita, Kira Ryhti, and Liisa Kulmala. Temperature and moisture dependence of daily growth of scots pine (pinus sylvestris l.) roots in southern finland. *Tree Physiology*, 40(2):272–283, 2020.

Alf Ekblad, Håkan Wallander, Douglas L Godbold, C Cruz, D Johnson, P Baldrian, RG Björk, Daniel Epron, B Kieliszewska-Rokicka, R Kjøller, et al. The production and turnover of extramatrical mycelium of ectomycorrhizal fungi in forest soils: role in carbon cycling. *Plant and Soil*, 366(1):1–27, 2013.

Oskar Franklin, Torgny Näsholm, Peter Högberg, and Mona N Högberg. Forests trapped in nitrogen limitation–an ecological market perspective on ectomycorrhizal symbiosis. *New Phytologist*, 203(2):657–666, 2014.

Heljä-Sisko Helmisaari, John Derome, Pekka Nöjd, and Mikko Kukkola. Fine root biomass in relation to site and stand characteristics in norway spruce and scots pine stands. *Tree Physiology*, 27(10):1493–1504, 2007.

Erik A Hobbie. Carbon allocation to ectomycorrhizal fungi correlates with belowground allocation in culture studies. *Ecology*, 87(3):563–569, 2006.

Mona N Högberg, Maria JI Briones, Sonja G Keel, Daniel B Metcalfe, Catherine Campbell, Andrew J Midwood, Barry Thornton, Vaughan Hurry, Sune Linder, Torgny Näsholm, et al. Quantification of effects of season and nitrogen supply on tree below-ground carbon transfer to ectomycorrhizal fungi and other soil organisms in a boreal pine forest. *New Phytologist*, 187(2): 485–493, 2010.

Teemu Hölttä, Miika Kurppa, and Eero Nikinmaa. Scaling of xylem and phloem transport capacity and resource usage with tree size. *Frontiers in plant science*, 4:496, 2013.

Torsten Ingestad, Agneta S Arveby, and Monika Käfar. The influence of ectomycorrhiza on nitrogen nutrition and growth of pinus sylvestris seedlings. *Physiologia Plantarum*, 68(4): 575–582, 1986.

Nancy Collins Johnson, Jason D Hoeksema, James D Bever, V Bala Chaudhary, Catherine Gehring, John Klironomos, Roger Koide, R Michael Miller, John Moore, Peter Moutoglis, et al. From lilliput to brobdingnag: extending models of mycorrhizal function across scales. *Bioscience*, 56(11):889–900, 2006.

Jaideep Joshi, Benjamin D Stocker, Florian Hofhansl, Shuangxi Zhou, Ulf Dieckmann, and Iain Colin Prentice. Towards a unified theory of plant photosynthesis and hydraulics. *Nature Plants*, 8(11):1304–1316, 2022.

Samuli Junttila, Teemu Hölttä, Ninni Saarinen, Ville Kankare, Tuomas Yrttimaa, Juha Hyyppä, and Mikko Vastaranta. Close-range hyperspectral spectroscopy reveals leaf water content dynamics. *Remote Sensing of Environment*, 277:113071, 2022.

Pirjo Kainulainen and JK Holopainen. Concentrations of secondary compounds in scots pine needles at different stages of decomposition. *Soil Biology and Biochemistry*, 34(1):37–42, 2002.

Deliang Kong, Junjian Wang, Huifang Wu, Oscar J Valverde-Barrantes, Ruili Wang, Hui Zeng, Paul Kardol, Haiyan Zhang, and Yulong Feng. Nonlinearity of root trait relationships and the root economics spectrum. *Nature communications*, 10(1):2203, 2019.

J. F. J. Korhonen, M. Pihlatie, J. Pumpanen, H. Aaltonen, P. Hari, J. Levula, A.-J. Kieloaho, E. Nikinmaa, T. Vesala, and H. Ilvesniemi. Nitrogen balance of a boreal scots pine forest. *Biogeosciences*, 10:1083–1095, 2013.

Kris R Kramer-Walter, Peter J Bellingham, Timothy R Millar, Rob D Smissen, Sarah J Richardson, and Daniel C Laughlin. Root traits are multidimensional: specific root length is independent from root tissue density and the plant economic spectrum. *Journal of Ecology*, 104 (5):1299–1310, 2016.

Björn D Lindahl, Katarina Ihrmark, Johanna Boberg, Susan E Trumbore, Peter Högberg, Jan Stenlid, and Roger D Finlay. Spatial separation of litter decomposition and mycorrhizal nitrogen uptake in a boreal forest. *New phytologist*, 173(3):611–620, 2007.

Björn D Lindahl, Wietse De Boer, and Roger D Finlay. Disruption of root carbon transport into forest humus stimulates fungal opportunists at the expense of mycorrhizal fungi. *The ISME journal*, 4(7):872–881, 2010.

Björn D Lindahl, Julia Kyaschenko, Kerstin Varenius, Karina E Clemmensen, Anders Dahlberg, Erik Karltun, and Johan Stendahl. A group of ectomycorrhizal fungi restricts organic matter accumulation in boreal forest. *Ecology Letters*, 24(7):1341–1351, 2021.

Annikki Mäkelä. Acclimation in dynamic models based on structural relationships. *Functional Ecology*, 13(2):145–156, 1999.

Annikki Mäkelä, Minna Pulkkinen, Pasi Kolari, Fredrik Lagergren, Paul Berbigier, Anders Lindroth, Denis Loustau, Eero Nikinmaa, Timo Vesala, and Pertti Hari. Developing an empirical model of stand gpp with the lue approach: analysis of eddy covariance data at five contrasting conifer sites in europe. *Global change biology*, 14(1):92–108, 2008.

Horst Marschner, Max Häussling, and Eckhard George. Ammonium and nitrate uptake rates and rhizosphere ph in non-mycorrhizal roots of norway spruce [picea abies (l.) karst.]. *Trees*, 5: 14–21, 1991.

ROSS E McMURTRIE. Relationship of forest productivity to nutrient and carbon supply—a modeling analysis. *Tree Physiology*, 9(1-2):87–99, 1991.

Maurizio Mencuccini and Lorenzo Bonosi. Leaf/sapwood area ratios in scots pine show acclimation across europe. *Canadian Journal of Forest Research*, 31(3):442–456, 2001.

Astrid Meyer, Rüdiger Grote, Andrea Polle, and Klaus Butterbach-Bahl. Simulating mycorrhiza contribution to forest c-and n cycling-the mycofon model. *Plant and soil*, 327:493–517, 2010.

F Minunno, Mikko Peltoniemi, Samuli Launiainen, M Aurela, A Lindroth, A Lohila, I Mammarella, K Minkkinen, and A Mäkelä. Calibration and validation of a semi-empirical flux ecosystem model for coniferous forests in the boreal region. *Ecological Modelling*, 341:37–52, 2016.

Torgny Näsholm, Knut Kielland, and Ulrika Ganeteg. Uptake of organic nitrogen by plants. *New phytologist*, 182(1):31–48, 2009.

Torgny Näsholm, Peter Högberg, Oskar Franklin, Daniel Metcalfe, Sonja G Keel, Catherine Campbell, Vaughan Hurry, Sune Linder, and Mona N Högberg. Are ectomycorrhizal fungi alleviating or aggravating nitrogen limitation of tree growth in boreal forests? *New Phytologist*, 198(1):214–221, 2013.

Eero Nikinmaa, Teemu Hölttä, Pertti Hari, Pasi Kolari, Annikki Mäkelä, Sanna Sevanto, and Timo Vesala. Assimilate transport in phloem sets conditions for leaf gas exchange. *Plant, Cell & Environment*, 36(3):655–669, 2013.

Erling Ögren. Relationship between temperature, respiratory loss of sugar and premature dehardening in dormant scots pine seedlings. *Tree Physiology*, 17(1):47–51, 1997.

Ivika Ostonen, Krista Lõhmus, Heljä-Sisko Helmisaari, Jaak Truu, and Signe Meel. Fine root morphological adaptations in scots pine, norway spruce and silver birch along a latitudinal gradient in boreal forests. *Tree Physiology*, 27(11):1627–1634, 2007.

Olusegun Ayodeji Oyewole, Erich Inselsbacher, Torgny Näsholm, and Sandra Jämtgård. Incorporating mass flow strongly promotes n flux rates in boreal forest soils. *Soil Biology and Biochemistry*, 114:263–269, 2017.

S Palmroth and P Hari. Evaluation of the importance of acclimation of needle structure, photosynthesis, and respiration to available photosynthetically active radiation in a scots pine canopy. *Canadian Journal of Forest Research*, 31(7):1235–1243, 2001.

Mikko Peltoniemi, Minna Pulkkinen, Mika Aurela, Jukka Pumpanen, Pasi Kolari, and Annikki Mäkelä. A semi-empirical model of boreal-forest gross primary production, evapotranspiration, and soil water-calibration and sensitivity analysis. *Luke Open Documents*, 2015.

Nazia Perveen, Sébastien Barot, Gaël Alvarez, Katja Klumpp, Raphael Martin, Alain Rapaport, Damien Herfurth, Frédérique Louault, and Sébastien Fontaine. Priming effect and microbial diversity in ecosystem functioning and response to global change: a modeling approach using the symphony model. *Global change biology*, 20(4):1174–1190, 2014.

Jaakko Repola. Biomass equations for scots pine and norway spruce in finland. 2009.

Kira Ryhti, Liisa Kulmala, Jukka Pumpanen, Jarkko Isotalo, Mari Pihlatie, Heljä-Sisko Helmisaari, Jaana Leppälammi-Kujansuu, Antti-Jussi Kieloaho, Jaana Bäck, and Jussi Heinonsalo. Partitioning of forest floor co2 emissions reveals the belowground interactions between different plant groups in a scots pine stand in southern finland. *Agricultural and Forest Meteorology*, 297:108266, 2021.

Kira Ryhti et al. Belowground carbon dynamics in scots pine stands. *Dissertationes Forestales*, 2022.

Pauliina Schiestl-Aalto, Liisa Kulmala, Harri Mäkinen, Eero Nikinmaa, and Annikki Mäkelä. Cassia–a dynamic model for predicting intra-annual sink demand and interannual growth variation in s cots pine. *New Phytologist*, 206(2):647–659, 2015.

Pauliina Schiestl-Aalto, Kira Ryhti, Annikki Mäkelä, Mikko Peltoniemi, Jaana Bäck, and Liisa Kulmala. Analysis of the nsc storage dynamics in tree organs reveals the allocation to belowground symbionts in the framework of whole tree carbon balance. *Frontiers in Forests and Global Change*, 2:17, 2019.

Sally E Smith and David J Read. *Mycorrhizal symbiosis*. Academic press, 2010.

Or Sperling, Tamir Kamai, Aude Tixier, Anna Davidson, Katherine Jarvis-Shean, Eran Raveh, Ted M DeJong, and Maciej A Zwieniecki. Predicting bloom dates by temperature mediated kinetics of carbohydrate metabolism in deciduous trees. *Agricultural and Forest Meteorology*, 276:107643, 2019.

Erica Sterkenburg, Adam Bahr, Mikael Brandström Durling, Karina E Clemmensen, and Björn D Lindahl. Changes in fungal communities along a boreal forest soil fertility gradient. *New Phytologist*, 207(4):1145–1158, 2015.

Benjamin N Sulman, Richard P Phillips, A Christopher Oishi, Elena Shevliakova, and Stephen W Pacala. Microbe-driven turnover offsets mineral-mediated storage of soil carbon under elevated co2. *Nature Climate Change*, 4(12):1099–1102, 2014.

Sanna Susiluoto, Emmi Hilasvuori, and Frank Berninger. Testing the growth limitation hypothesis for subarctic scots pine. *Journal of Ecology*, 98(5):1186–1195, 2010.

Xianglin Tian, Francesco Minunno, Pauliina Schiestl-Aalto, Jinshu Chi, Peng Zhao, Matthias Peichl, John Marshall, Torgny Näsholm, Hyungwoo Lim, Mikko Peltoniemi, et al. Disaggregating the effects of nitrogen addition on gross primary production in a boreal scots pine forest. *Agricultural and Forest Meteorology*, 301:108337, 2021.

Oscar J Valverde-Barrantes, Grégoire T Freschet, Catherine Roumet, and Christopher B Blackwood. A worldview of root traits: the influence of ancestry, growth form, climate and mycorrhizal association on the functional trait variation of fine-root tissues in seed plants. *New Phytologist*, 215(4):1562–1573, 2017.

Anouk Van't Padje, Gijsbert DA Werner, and E Toby Kiers. Mycorrhizal fungi control phosphorus value in trade symbiosis with host roots when exposed to abrupt 'crashes' and 'booms' of resource availability. *New Phytologist*, 229(5):2933–2944, 2021.

T Wallenda and DJ Read. Kinetics of amino acid uptake by ectomycorrhizal roots. *Plant, Cell & Environment*, 22(2):179–187, 1999.

Natalia Wojciechowska, Katarzyna Marzec-Schmidt, Ewa M. Kalemba, Agnieszka Ludwików, and Agnieszka Bagniewska-Zadworna. Seasonal senescence of leaves and roots of populus trichocarpa—is the scenario the same or different? *Tree Physiology*, 40(8):987–1000, 2020.